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CHAPTER 30

Microorganisms in Terrestrial Environments



Terrestrial plants and filamentous fungi have developed long-term relationships that benefit both partners. The tips of pine tree roots are usually surrounded by dense fungal sheaths that are part of a hyphal network which extends out into the soil. The plant supplies organic matter to maintain the fungus, and the fungus, in turn, provides the plant with nutrients and water.

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Concepts

1. Terrestrial [Latin *terra*, earth] environments are dominated by inert solid materials. Organic substances, including microorganisms, are usually a minor part of a soil.
2. Water, when present, will be predominantly in thin water films on particle surfaces. Microorganisms in these water films have good contact with gases in air, including oxygen.
3. Oxygen is much more accessible to microorganisms in soils because its flux into thin water films is not limited by a large continuous water bulk phase. The flux rate of oxygen in air is approximately 4000 times faster than in water.
4. Microorganisms growing in such higher oxygen flux environments have developed complex physical and physiological mechanisms to deal with potentially toxic oxygen species. Fungi even develop structures impermeable to oxygen.
5. Soils can have isolated water-saturated zones that become anaerobic. These may be considered as isolated aquatic environments surrounded by high-oxygen flux zones.
6. There is a great variety of terrestrial environments: dry, cold, temperate, tropical, and geothermally heated zones.
7. Insects, nematodes, and other soil animals interact with microorganisms to influence nutrient cycling and other processes.
8. The subsurface supports a varied and diverse microbial community. These microorganisms are maintained by nutrient sources such as materials leaching from the surface, decomposition of buried plant remains, and methane synthesis.
9. Fungi play important roles in the functioning of plants. They occur inside and on the surfaces of plants, and can have positive and negative effects. Most plant roots are infected with fungi.
10. Soil microorganisms interact with plants and the atmosphere. Critical microbial products are nitrous oxide and methane, both greenhouse gases. In addition, microorganisms produce chloromethane and cyanide, usually considered as anthropogenic pollutants.

They [the leaves] that waved so loftily, how contentedly, they return to dust again and are laid low, resigned to lie and decay at the foot of the tree and afford nourishment to new generations of their kind, as well as to flutter on high!

—Henry D. Thoreau

Soils are dynamic and develop over time. This process may take decades and even centuries; the organic matter in soils can be thousands of years old. Because of changes in plant growth, temperature, rainfall, disturbance and erosion, a soil that has taken hundreds of years to form can be quickly degraded if the microbial community is activated. For example, this can occur when a bog soil is drained, which increases oxygen access to the accumulated organic matter. In most soils the major producers of organic matter are the vascular plants, although algae, cyanobacteria, and photosynthetic bacteria also can contribute to these processes, especially in desert crust environments.

Soil is the habitat for a variety of organisms, including bacteria, fungi, protozoa, insects, nematodes, worms, and many other animals. Viruses also are present in soils. This complex

biological community contributes to the formation, maintenance, and in some situations, the degradation and disappearance of soils.

30.1 Soils as an Environment for Microorganisms

Soils have been formed, and continue to be formed, in a wide variety of environments. These environments range from Arctic tundra regions, where approximately 11% of the world's soil carbon pool is stored, to Antarctic dry valleys, where there are no vascular plants. In addition, deeper subsurface zones, where plant roots and their products cannot penetrate, also have microbial communities. Microbial activities in these environments can lead to the formation of minerals such as dolomite; microbial activity also occurs in deep continental oil reservoirs, in stones, and even in rocky outcrops. These microbes are dependent on energy sources from algae, nutrients in rainfall, and dust.

Most soils are dominated by inorganic geological materials, which are modified by the biotic community, including the microorganisms and plants, to form soils. A major characteristic of soils is that they are typically not water saturated, making it possible for oxygen to penetrate into the passages and pores. Soils also can contain isolated regions which are water saturated, and these isolated regions become “mini” aquatic environments.

An important defining characteristic of a soil, from a microbial viewpoint, is that the microorganisms are in close physical contact with oxygen; they are located in thin water films on the particle surfaces where oxygen is present at high levels and can be easily replenished from the gaseous phase. When microorganisms use oxygen, it can be replenished rapidly by diffusion, thus maintaining the microbes under aerobic conditions. Oxygen diffusion through air in the soil occurs about 4,000 times faster than it does through water (**figure 30.1**). As shown in this figure, the oxygen concentrations and flux rates in pores and channels is high, whereas within water-filled zones the oxygen flux rate is much lower. As an example, particles as small as about 2.0 mm can be aerobic on the outside and anaerobic on the inside. Even in oxygen-saturated zones, microbial-scale aquatic environments can be created that are “hot spots” for anaerobic processes.

Depending on the physical characteristics of the soil, rainfall or irrigation may rapidly change a soil from being ideal, in terms of having reoxygenating thin water films, to an environment with isolated pockets of water, which are “miniaquatic” habitats. If this process of flooding continues, a water-logged soil can be created that is more like a lake sediment.

Shifts in water content and gas fluxes also affect the concentrations of CO₂, CO, and other gases present in the soil atmosphere, as noted in **table 30.1**. These changes will be accentuated in the smaller pores where many bacteria are found. At lower depths less oxygen is available, especially in wetter, less permeable soils. Another factor that affects the levels of oxygen and CO₂ in a soil are plant roots. The roots of plants growing in normal aerated soils also consume oxygen and release CO₂, influencing the concentrations of these gases in the root environment.

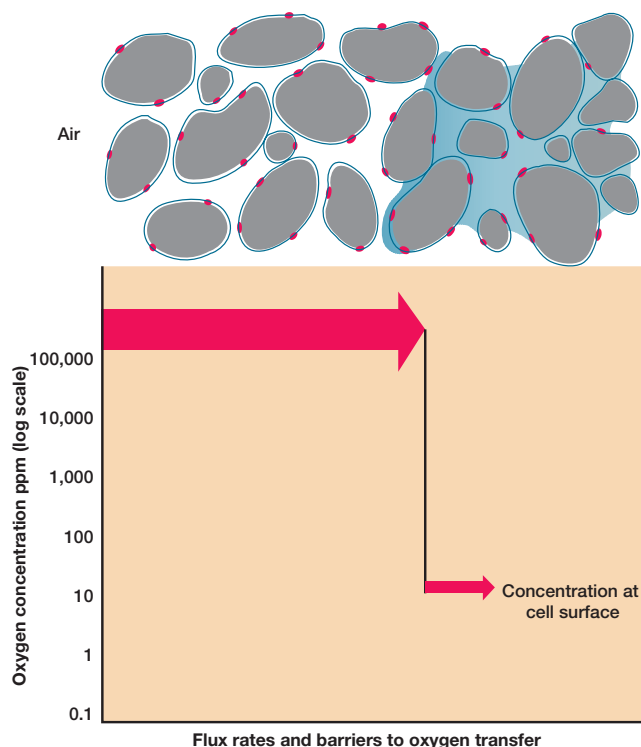


Figure 30.1 Oxygen Concentrations and Fluxes in a Soil.

Microorganisms in thin water films on the surface of soil particles have good access to oxygen. In comparison, microbes in isolated water volumes have limited oxygen fluxes, creating miniaquatic environments.

1. Why are Arctic tundra soils important in terms of carbon storage?
2. Contrast differences in oxygen flux rates and concentrations in a soil with those in a miniaquatic environment.
3. How do the concentrations of oxygen and carbon dioxide differ between the atmosphere and the soil interior?

30.2 Microorganisms in the Soil Environment

If we look at a soil in greater detail (**figure 30.2**), the bacteria and fungi use different functional strategies to take advantage of this complex physical matrix. Most soil bacteria are located on the surfaces of soil particles and require water and nutrients that must be located in their immediate vicinity. Bacteria are found most frequently on surfaces within smaller soil pores (2 to 6 μm in diameter). Here they are probably less liable to be eaten by protozoa, unlike bacteria that are located on the exposed outer surface of a sand grain or organic matter particle.

Table 30.1 Concentrations of Oxygen and Carbon Dioxide in the Atmosphere of a Tropical Soil under Wet and Dry Conditions

Soil Depth (cm)	Oxygen Content (%)		Carbon Dioxide Content (%)	
	Wet	Dry	Wet	Dry
10	13.7	20.7	6.5	0.5
25	12.7	19.8	8.5	1.2
45	12.2	18.8	9.7	2.1
90	7.6	17.3	10.0	3.7
120	7.8	16.4	9.6	5.1

From E. W. Russell, *Soil Conditions and Plant Growth*, 10th edition. Copyright © 1973 Longman Group Limited, Essex, United Kingdom. Reprinted by permission.

Note: Normal air contains approximately 21% oxygen and 0.035% carbon dioxide.

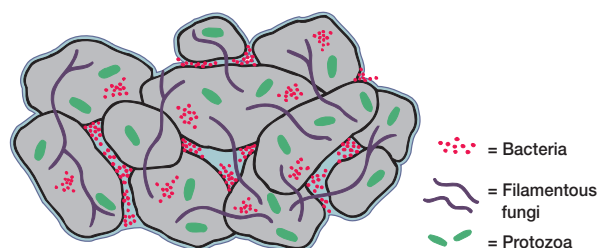


Figure 30.2 The Microenvironment—The World of Microorganisms in Soil.

Bacteria tend to be present as isolated microcolonies on surfaces and in pores, which are covered by thin water films. Filamentous fungi are able to grow on and between these aggregated particles, or peds. Protozoa move in water films and graze on bacteria, especially when they are not protected in soil pores.

Terrestrial filamentous fungi bridge across open areas between soil particles or aggregates called **peds**, and are exposed to high levels of oxygen (figure 30.2). These fungi will tend to darken and form oxygen-impermeable structures including sclerotia and hyphal cords. This is particularly important for the functioning of basidiomycetes, which form such structures as an oxygen sealing mechanism. Within these structures, the filamentous fungi move nutrients and water over great distances, including across air spaces, a unique part of their functional strategy. These oxidatively polymerized, oxygen-impermeable hyphal boundaries do not usually occur in fungi growing in aquatic environments. **Basidiomycetes** (pp. 561–64)

A wide variety of insects and animals also are present in soils, and these often use the fungi and bacteria as food sources, as well as processing plant residues. The earthworms, with their ability to mix and ingest soils, add bacteria and enzymes from their intestine to their soil casts, a process that has major effects on the soil structure and the soil microbial community. Earth-

Table 30.2 Easily Cultured Gram-Positive
Irregular Branching and Filamentous
Bacteria Common in Soils

Bacterial Group	Representative Genera	Comments and Characteristics
Coryneforms	<i>Arthrobacter</i>	Rod-coccus cycle
	<i>Cellulomonas</i>	Important in degradation of cellulose
Mycobacteria	<i>Corynebacterium</i>	Club-shaped cells
	<i>Mycobacterium</i>	Acid-fast
Nocardioforms	<i>Nocardia</i>	Rudimentary branching
Actinomycetes	<i>Streptomyces</i>	Aerobic filamentous bacteria
	<i>Thermoactinomyces</i>	Higher temperature growth

worms, along with many other soil insects and animals, use filamentous fungi as a food source, and this can result in decreased fungal development in a soil. Earthworms also assist in mixing soil organic materials, creating the deep soils found in grasslands. In forest areas, which lack as many earthworms, a more distinct organic matter layer will be formed that is separated from the underlying inorganic layer. [The fungi \(chapter 25\)](#)

The microbial populations in soils can be very high. In a surface soil the bacterial population can approach 10^8 to 10^9 cells per gram dry weight of soil as measured microscopically. Fungi can be present at up to several hundred meters of hyphae per gram of soil. We tend to think that soil fungi are small like the mushrooms sprouting from our lawns. This is natural because most of the fungal thallus lies beneath the soil surface, but such a view sometimes is quite inaccurate. A case in point is the fungus *Armillaria bulbosa*, which lives associated with tree roots in hardwood forests. An individual *Armillaria* clone that covers about 30 acres has been discovered in the Upper Peninsula of Michigan. It is estimated to weigh a minimum of 100 tons (an adult blue whale may weigh 150 tons) and be at least 1,500 years old. Thus some fungal mycelia are among the largest and most ancient living organisms on Earth.

It is important to remember, when discussing soils and their microorganisms, that only a minor portion (approximately 10%) of the microscopically observable organisms making up this biomass have been cultured. In terms of biotechnology and basic ecology, the microorganisms that have not been cultured may provide a valuable genetic resource for basic and applied research. We are gradually learning more about these uncultured microorganisms through the use of molecular techniques. For example, novel groups of the *Crenarchaeota* (see pp. 456–58) have been discovered in forest soil and the ocean by extracting microbial DNA and amplifying it with the polymerase chain reaction. Examination of soils from different areas of the world continues to yield surprises. Recent molecular studies of Siberian tundra soils have uncovered considerable unexplored microbial diversity, as most of the recovered bacteria are not related to any known species. Microorganisms are present and prolific in subsurface environments, including oil

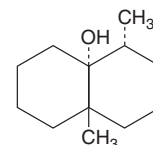
*trans*-1,10-dimethyl-*trans*-9-decalol

Figure 30.3 Geosmin. The structure of geosmin, the major chemical produced by actinomycetes and cyanobacteria that gives soils their characteristic odor.

reservoirs. Hyperthermophilic archaea have been found in such harsh subsurface environments and are probably indigenous to these poorly understood regions of our world.

The gram-positive bacteria, which show varied degrees of branching and mycelial development, are an important and less studied part of the soil microbial community. They include the coryneforms, the nocardioforms, and the true filamentous bacteria or actinomycetes ([table 30.2](#)). These bacteria play a major role in the degradation of hydrocarbons, older plant materials, and soil humus. In addition, some members of these groups actively degrade pesticides. The filamentous actinomycetes, primarily of the genus *Streptomyces*, produce an odor-causing compound called **geosmin**, which gives soils their characteristic earthy odor ([figure 30.3](#)).

[Actinobacteria \(chapter 24\)](#)

As with aquatic environments (see [chapter 29](#)), microorganisms are constantly being added to soils from waters, wind, dust, plants, and animal sources. Most of these added microorganisms will not survive, either being outcompeted by indigenous microbes or being consumed by predators such as the protozoa.

The microbial community in soil makes important contributions to biogeochemical cycling and the carbon, nitrogen, sulfur, iron, and manganese cycles (see [chapter 28](#)). Because the soil is primarily an oxidized environment, the inorganic forms of these elements will tend to be in the oxidized state. If there are localized water-saturated, lower oxygen-flux environments, the biogeochemical cycles will shift toward reduced species. [Biogeochemical cycles, \(section 28.3\)](#)

Soil microorganisms can be categorized with respect to organic matter processing on the basis of both (1) their preference for either easily available or more resistant substrates and (2) the substrate concentrations they require. At higher nutrient-level levels, some microorganisms such as *Pseudomonas* respond rapidly to the addition of easily usable substrates such as sugars and amino acids. Indigenous forms tend to use native organic matter to a greater extent. These include the genus *Arthrobacter* and many soil actinomycetes. A less understood part of the microbial community grows in oligotrophic environments, defined as those that contain less than 15 mg/liter of organic matter. Physiological and genetic studies on microbial responses in such low-nutrient environments are under way. As discussed in this book, microbes have many strategies to deal with survival in such low-nutrient environments, and this presents special technical problems when studying them. [Arthrobacter \(pp. 542–43\)](#)

Soil insects and other animals such as earthworms also contribute to organic matter transformations in soils. These organisms carry out decomposition, often leading to the release of minerals, and physically “reducing” the size of organic particles such as plant litter. This increases the surface area and makes organic materials more available for use by bacteria and fungi. These organisms also mix substrates with their internal gut microflora and enzymes, and this contributes substantially to decomposition.

Protozoa also can influence nutrient cycling by feeding on “palatable” microorganisms. This process of **microbivory**, or use of microorganisms as a food source, results in higher rates of nitrogen and phosphorus mineralization, thus increasing the availability of nutrients for plant growth. [Protozoa \(chapter 27\)](#)

A significant part of the biological activity of soils arises from enzymes released by plants, insects, and other animals, and from lysed microorganisms. These free enzymes contribute to many hydrolytic degradation reactions, such as proteolysis; catalase and peroxidase activities also have been detected. Apparently these free enzymes associate with clays and humic materials, which helps to protect them from denaturation and microbial degradation.

A microbial loop regenerates nutrients in soils as it does in aquatic environments. In this process carbon and nutrients fixed by plants will be turned over rapidly by bacteria and predatory protozoa and will not be available for transfer to higher trophic levels in the food web. Protozoa, predominantly the naked amoebae that glide over surfaces, are the major protozoan predators. In waters the major predators are ciliates and flagellates.

1. What are the differences in preferred soil habitats between bacteria and filamentous fungi?
2. How can earthworms, nematodes, and insects influence microbial communities?
3. What types of archaeans have been detected in soils?
4. Does the microbial loop function in soils?

30.3 Microorganisms and the Formation of Different Soils

Soils form under various environmental conditions. When newly exposed geologic materials begin to weather, as after a volcanic event or a simple soil disturbance, microbial colonization occurs. If only subsurface materials are available, phosphorus may be present, but nitrogen and carbon must be imported by physical or biological processes. Under these circumstances many cyanobacteria, which can fix atmospheric nitrogen and carbon, are active in pioneer-stage nutrient accumulation.

Once they are formed, most soils are rich sources of nutrients. Nutrients are found in organic matter, microorganisms, soil insects, and other animals. Plants grow, senesce, and die—and at each of these phases, they provide nutrients for soil organisms. Different plant parts vary in their nutrient content and biomasses. In addition, the turnover times for the various plant parts are quite different. The components in the plant-soil system with the lowest carbon-nitrogen ratios (most nutrient-rich) are soil organic

matter, microorganisms, soil insects, and other soil animals. The soil organic matter contains the greatest portion of the carbon and nitrogen in a typical soil, but with its slow turnover time (100 to 1,000 years and longer), most of this nutrient resource is not immediately available for plant or microbial use. The major types of soils that can be formed are shown in **figure 30.4**.

Tropical and Temperate Region Soils

In moist tropical soils, with their higher mean temperatures, organic matter is decomposed very quickly, and the mobile inorganic nutrients can be leached out of the surface soil environment, causing a rapid loss of fertility. To limit nutrient loss, many tropical plants have root systems that penetrate the rapidly decomposing litter layer. As soon as organic material and minerals are released during decomposition, the roots can take them up to avoid losses in leaching. Thus it is possible to “recycle” nutrients before they are lost with water movement through the soil (figure 30.4a). With deforestation, the nutrients are not recycled, leading to their loss from the soil and decreased soil fertility.

Tropical plant-soil communities are often used in **slash-and-burn agriculture**. The vegetation on a site is chopped down and burned to release the trapped nutrients. For a few years, until the minerals are washed from the soils, crops can be grown. When the minerals are lost from these low organic matter soils, the farmer must move to a new area and start over by again cutting and burning the native plant community. This cycle of slash-and-burn agriculture is stable if there is sufficient time for the plant community to regenerate before it is again cut and burned. If the cycle is too short, as can occur with overpopulation, rapid and almost irreversible degradation of the soil can occur.

In many temperate region soils, in contrast, the decomposition rates are less than that of primary production, leading to litter accumulation. Deep root penetration in temperate grasslands results in the formation of fertile soils, which provide a valuable resource for the growth of crops in intensive agriculture (figure 30.4b).

The soils in many cooler coniferous forest environments suffer from an excessive accumulation of organic matter as plant litter (figure 30.4c). In winter, when moisture is available, the soils are cool, and this limits decomposition. In summer, when the soils are warm, water is not as available for decomposition. Organic acids are produced in the cool, moist litter layer, and they leach into the underlying soil. These acids solubilize soil components such as aluminum and iron, and a bleached zone may form. Litter continues to accumulate, and fire becomes the major means by which nutrient cycling is maintained. Controlled burns are becoming a more important part of environmental management in this type of plant-soil system.

Bog soils provide a unique set of conditions for microbial growth (figure 30.4d). In these soils the decomposition rate is slowed by the waterlogged, predominantly anoxic conditions, which lead to peat accumulation. When such areas are drained, they become more aerobic and the soil organic matter is degraded, resulting in soil subsidence. Under aerobic conditions the lignin-cellulose complexes of the accumulated organic matter are more susceptible to decomposition by the filamentous fungi.

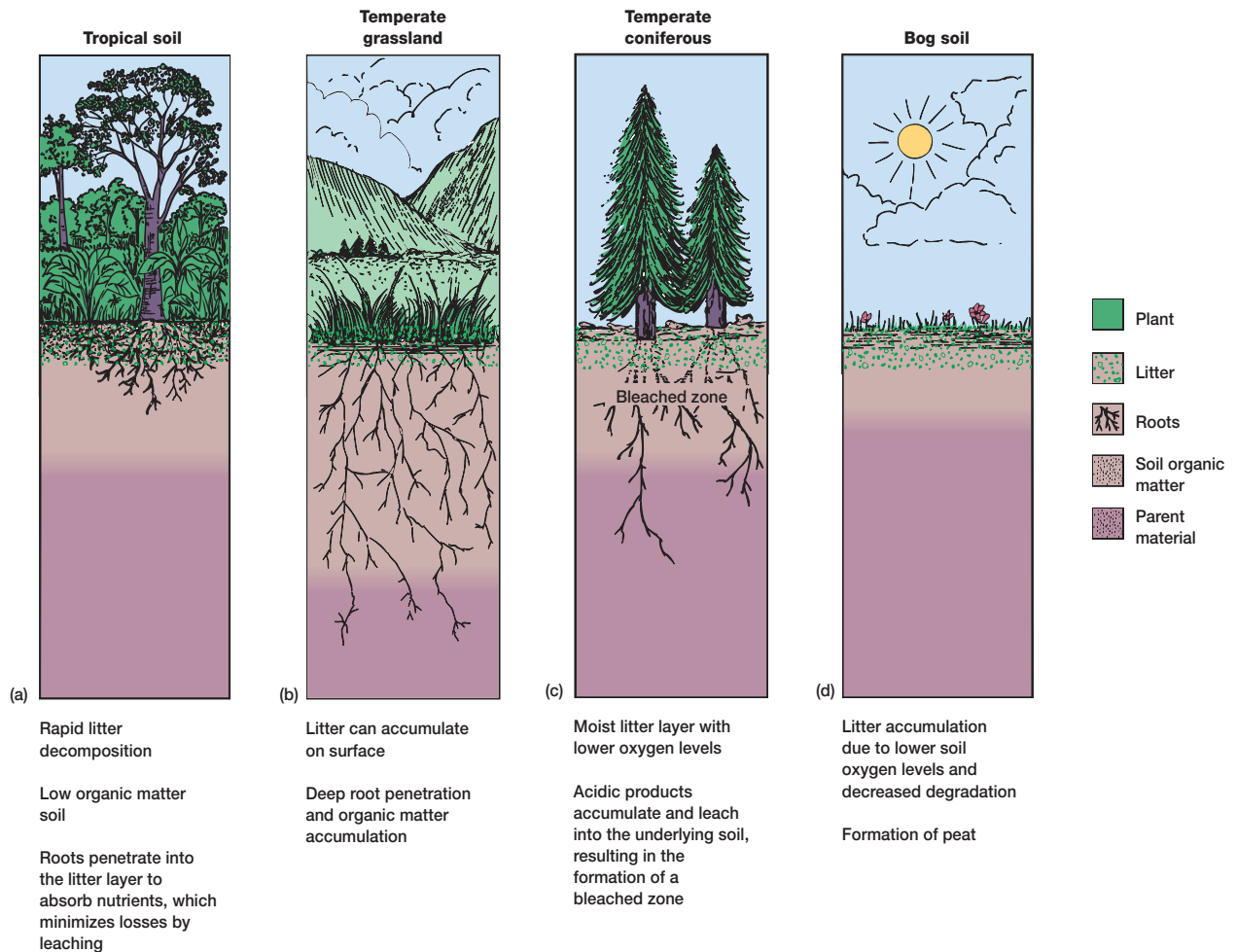


Figure 30.4 Examples of Tropical and Temperate Region Plant-Soil Systems. Climate, parent material, plants, topography, and microorganisms interact over time to form different plant-soil systems. In these figures the characteristics of (a) tropical, (b) temperate grassland, (c) temperate coniferous forest, and (d) bog soils are illustrated.

Cold Moist Area Soils

Soils in cold environments, whether in Arctic, Antarctic, or alpine regions, are of extreme interest because of their wide distribution and impacts on global-level processes. The colder mean soil temperatures at these sites decrease the rates of both decomposition and plant growth. In these cases soil organic matter accumulates, and plant growth can become limited due to the immobilization of nutrients in soil organic matter. Often, below the plant growth zone, these soils are permanently frozen. These permafrost soils hold 11% of the Earth's soil carbon and 95% of its organically bound nutrients. These soils are very sensitive to physical disturbance and pollution, and the widespread exploration of such areas for oil and minerals can have long-term effects on their structure and function. When plants are covered by snow, plant diseases are caused by **snow molds**, which can grow and attack

plants at -5°C . These fungi produce special antifreeze proteins (AFP) that enable them to function under these harsh conditions.

In water-saturated bog areas, bacteria are more important than fungi in decomposition processes, and there is decreased degradation of lignified materials. As in other soils, the nutrient cycling processes of nitrification, denitrification, nitrogen fixation, and methane synthesis and utilization, although occurring at slower rates, can have major impacts on global gaseous cycles.

[Lignin degradation \(pp. 613–14\)](#)

Desert Soils

Soils of hot and cold arid and semiarid deserts are dependent on periodic and infrequent rainfall. When these rainfalls occur, water can puddle in low areas and be retained on the soil surface by microbial communities called **desert crusts**. These consist of



Figure 30.5 A Desert Crust as Observed with the Scanning Electron Microscope. The crust has been disturbed to show the extracellular sheaths and filaments of the cyanobacterium *Microcoleus vaginatus*. The sand grains are linked by these filamentous growths, creating a unique ecological structure.

cyanobacteria and associated commensalistic microbes, including *Anabaena*, *Microcoleus*, *Nostoc*, and *Scytonema*. The depth of the photosynthetic layer is perhaps 1 mm, and the cyanobacterial filaments and slimes link the sand particles (**figure 30.5**), which change the surface soil albedo, water infiltration rate, and susceptibility to erosion. These crusts are quite fragile, and vehicle damage can be evident for decades. After a rainfall, nitrogen fixation will begin within approximately 25 to 30 hours, and when the rain has evaporated or drained into the soil, the crust will dry up and nitrogen will be released for use by other microorganisms and the plant community.

Geologically Heated Hyperthermal Soils

Geologically heated soils are found in such areas as Iceland and the Kamchatka peninsula in eastern Russia, and such heated soils also occur at many mining waste sites. An important microorganism found in heated mining wastes is *Thermoplasma*. These soils are populated by bacterial and archaeal procaryotes, many of which are chemolithoautotrophs. A wide variety of chemoorganotrophic genera also are found in these environments; these include the aerobes *Thermomicrobium*, *Thermoleophilum*, and also the anaerobes *Thermosipho* and *Thermotoga*. Such geothermal soils have been of great interest as a source of new microbes to use in biotechnology (see p. 626), and the search for new, unique microorganisms in these areas is intensifying all over the world. *Thermoplasma* (p. 463)

1. Characterize each major soil type discussed in this section in terms of the balance between primary production and organic matter decomposition.
2. What is unique about bogs in terms of organic matter degradation?
3. Why can nutrients become limiting for plants and microorganisms in tundra soils?
4. Describe desert crusts. What types of microorganisms function in these unique environments?
5. What is slash-and-burn agriculture? Describe the roles of microorganisms in this process.
6. What unique microbial genera are found in geothermally heated soils?

30.4 Soil Microorganism Associations with Vascular Plants

Plants—the major source of organic matter on which most soil microorganisms are dependent—are heavily colonized with microorganisms. Microorganisms of various kinds are associated with the leaves, stems, flowers, seeds, and roots. The microbial community influences plants in many direct and indirect ways. The community includes microorganisms that grow on the plant surface, between cells on the surface of the plant, and within plant cells.

Microorganisms on the Outside of Plants

Plants are covered by microorganisms, on both the above- and belowground surfaces. These microbes usually are simple commensals; in some cases they can become parasitic or pathogenic, especially when the plant surface has been damaged.

Phyllosphere Microorganisms

A wide variety of microorganisms are found on and in the aerial surfaces of plants, called the **phyllosphere**. These include microorganisms that have complex interactions with the plant at various stages of development. The plant leaves and stems release organic compounds, and this can lead to massive development of microbes. The genera present on plant leaves and stems include *Sphingomonas*, which is especially equipped to survive with the high levels of UV irradiation occurring on these plant surfaces. This bacterium, also common in soils and waters, can occur at 10^8 cells per gram of plant tissue. *Sphingomonas* often represents a majority of the culturable species. Phyllosphere microorganisms play important roles in protection and possibly harm to the plant.

Rhizosphere and Rhizoplane Microorganisms

Plant roots release a wide variety of materials to their surrounding soil, including various alcohols, ethylene, sugars, amino and organic acids, vitamins, nucleotides, polysaccharides, and enzymes as shown in **table 30.3**. These materials create unique en-

Table 30.3 Compounds Excreted by
Microorganism-Free Wheat Roots

Volatile Compounds	Low-Molecular-Weight Compounds	High-Molecular-Weight Compounds
CO ₂	Sugars	Polysaccharides
Ethanol	Amino acids	Enzymes
Isobutanol	Vitamins	
Isoamyl alcohol	Organic acids	
Acetoin	Nucleotides	
Isobutyric acid		
Ethylene		

From J. W. Woldendorp, "The Rhizosphere as Part of the Plant-Soil System" in *Structure and Functioning of Plant Populations* (Amsterdam, Holland: Proceedings, Royal Dutch Academy of Sciences, Natural Sciences Section: 2d Series, 1978) 70:243.

vironments for the soil microorganisms. These environments include the **rhizosphere**, first described by Lorenz Hiltner in 1904, the volume of soil around the root influenced by the materials released from the root. The plant root surface, termed the **rhizoplane**, also provides a unique environment for microorganisms, as these gaseous, soluble, and particulate materials move from the plant to the soil. The rhizosphere and rhizoplane microorganisms not only increase their numbers when these newly available substrates become available, but their composition and function also change. Rhizosphere and rhizoplane microorganisms, as they respond to these released materials, also serve as labile sources of nutrients for other organisms, thus creating a soil microbial loop in addition to playing critical roles in organic matter synthesis and degradation. [Microbial loop \(chapter 29\)](#)

A wide range of bacteria in the rhizosphere can promote plant growth, orchestrated by rhizosphere bacteria that communicate with the plant using complex chemical signals. These chemical signal compounds, including auxins, gibberellins, glycolipids, and cytokinins, are only now beginning to be fully appreciated in terms of their biotechnological potential. Plant growth/promoting rhizobacteria include the genera *Pseudomonas* and *Achromobacter*. These can be added to the plant, even in the seed stage, if the bacteria have the required attachment surface proteins. The genes that control the expression of these attachment proteins are still being identified.

A critical process that occurs on the surface of the plant, and particularly in the root zone, is **associative nitrogen fixation**, in which the nitrogen-fixing microorganism is on the surface of the plant root, the rhizoplane (**figure 30.6**), as well as in the rhizosphere. This process is carried out by representatives of the genera *Azotobacter*, *Azospirillum* and *Acetobacter*. These bacteria contribute to nitrogen accumulation by tropical grasses. Recent evidence suggests that their major contribution may not be in nitrogen fixation but in production of growth-promoting hormones that increase root hair development and thus greater ability of the plant to take up nutrients. This is an area of research that is particularly important in tropical agricultural areas.

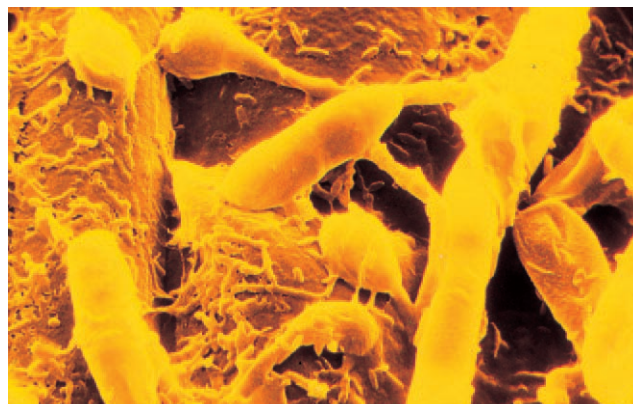


Figure 30.6 Root Surface Microorganisms. Plant roots release nutrients that allow intensive development of bacteria and fungi on and near the plant root surface, the rhizoplane. A scanning electron micrograph shows bacteria and fungi growing on a root surface.

1. Define the following terms: rhizosphere, rhizoplane, and associative nitrogen fixation.
2. What important genera are involved in associative nitrogen fixation?
3. What unique stresses face a microorganism on a leaf but not a microorganism in the soil?
4. What microbial genera are considered as plant growth-promoting rhizobacteria?

Microorganism Growth within Plants

In addition to the microorganisms growing on the surface of a plant, many interesting and important microbes grow within plants. These associations are dependent upon complex chemical signals, which indicates that these relationships are very ancient. It is important to emphasize that the mechanisms by which many of these plant-microbe interactions occur are also found in microorganism-animal interactions (*see chapter 31*).

Rhizobium

Several microbial genera are able to form nitrogen-fixing nodules with legumes. These include *Allorhizobium*, *Azorhizobium*, *Bradyrhizobium*, *Mesorhizobium*, *Sinorhizobium*, and *Rhizobium*. The mechanism of this interaction will be illustrated by work with *Rhizobium*.

The genus *Rhizobium* (*see p. 492*) is a prominent member of the rhizosphere community. This bacterium also can establish a symbiotic association with legumes and fix nitrogen for use by the plant. *Rhizobium* infects and nodulates specific legume hosts. The bacterium contains a large plasmid that encodes information that is not used when it grows as a free-living organism in the soil, but is vital for infection and nodulation of the susceptible host

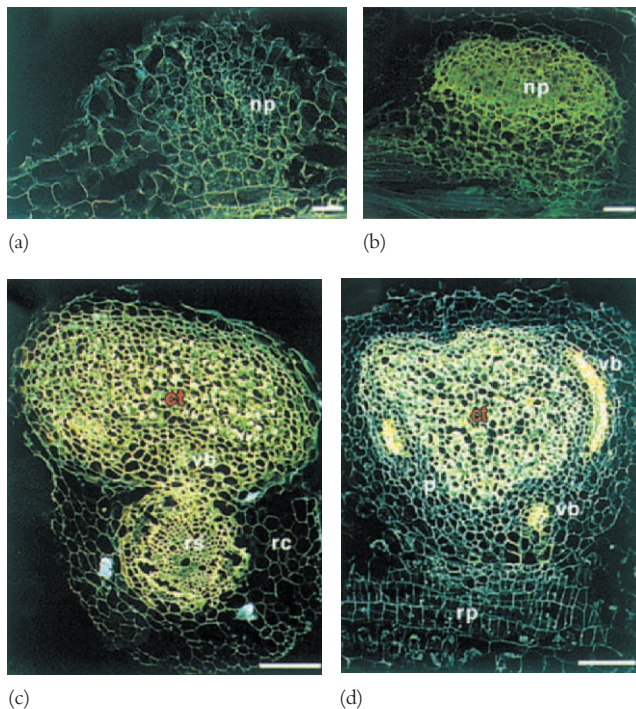


Figure 30.7 Controlled Initial Development of Legume Root Nodules Incited by *Rhizobium*. Expression of nodule inception (*nin*) activity is a critical step in infection thread formation and initiation of primordia: (a) early stage of primordial cell development with low levels of *nin* activity, (b) later higher-level expression of *nin* activity, (c) transverse nodule section showing extended *nin* expression, (d) fully differentiated nitrogen-fixing nodule with *nin* activity in central tissue area. In situ hybridization with probes used to detect activity. Abbreviations used: np, nodule primordium; ct, central tissue; rc, root cortex; rs, root stele; vb, vascular bundle. Bar = 100 μm .

plant. Plasmid-encoded genes also influence the range of host plants that *Rhizobium* can nodulate.

The *Rhizobium*-legume symbiosis is based on complex molecular interactions much like successive locks and doors that control this process. The *Rhizobium* infection process is controlled by the *bacA* gene that is required to establish the nodule. *Brucella abortus* has a similar *bacA* gene that is required to produce a chronic infection in animal hosts. Another similarity with pathogens is the *Lon* protease, which is important in *Rhizobium* and also influences capsule production in *E. coli*, *Klebsiella*, and *Erwinia*. Symbiotic development in legumes is controlled by a plant regulator protein that dedifferentiates root cortical cells and restarts cell division to establish nodule primordia within the invaded root. The primordia are invaded by *Rhizobium* using infection threads created by the plant. The establishment of the nodules involves a newly discovered plant gene called *nin*. Expression of this gene during nodule development has been detected by the use of in situ probes in legume root tissue (figure 30.7).

This complex infection process (figure 30.8), presented here in a simplified form, appears to involve a series of molecules produced by the host plant that lead to the exchange of recognition signals. Flavonoid inducers produced by the plant play a major role in this process by stimulating the *Rhizobium* to synthesize specific **Nod factors** that activate the host symbiotic processes necessary for root hair infection and nodule development (figure 30.8a–d). After bacterial attachment, the root hairs curl and the bacteria induce the plant to form an **infection thread** that grows down the root hair.

The *Rhizobium* then spreads within the infection thread into the underlying root cells as noted in figure 30.8e. At no time does the *Rhizobium* actually enter the plant cytoplasm while it is in the infection thread! When the bacteria are released from the infection thread into the host cell (figure 30.8g) the *Rhizobium* is enclosed by a plant-derived membrane, called the peribacteroid membrane, to form a **bacteroid** (figure 30.8h). Further growth and differentiation lead to the development of a nitrogen-fixing form, a structure called a **symbiosome** (figure 30.8h). At this point, specific nodule components such as leghemoglobin, which protect the nitrogen fixation enzymes from oxygen, are produced to complete the nodulation process (figure 30.8i). The final **root nodules** are shown in figure 30.8j.

The symbiosomes within mature root nodules are the site of nitrogen fixation. Within these nodules, the differentiated bacteroids reduce atmospheric N_2 and form ammonia (the primary product) and alanine; these compounds are released into the host plant cell, assimilated into various other nitrogen-containing organic compounds, and distributed throughout the plant. Because reduced nitrogen is the nutrient most commonly limiting plant growth, biological nitrogen fixation, as exemplified by the *Rhizobium*-legume symbiosis, is of major importance to agricultural productivity and the biogeochemical nitrogen cycle needed to sustain life on Earth.

After the *Rhizobium* cells have developed into the final differentiated bacteria, they no longer can revert back to bacteria capable of reproduction and have been lost to the “gene pool.” The fittest members of the bacterial community, which have achieved nodulation, thus have sacrificed their own reproduction! This “altruism in the rhizosphere,” a concept suggested by I. Olivieri and S. A. Frank, actually may promote increased root exudation and the maintenance of higher numbers of *Rhizobium* in the plant-root environment. The molecular biology of nitrogen fixation by *Rhizobium*, with the complex chemical communications involved, continues to be a subject of intense study around the world.

A major goal of biotechnology is to introduce nitrogen fixation genes into plants that do not normally form such associations. Recently it has been possible to produce modified lateral roots on nonlegumes such as rice, wheat, and oilseed rape; the roots are invaded by nitrogen-fixing bacteria. It appears that the infection begins with bacterial attachment to the root tips. Although these modified root structures have not yet been found to fix useful amounts of nitrogen, they do enhance rice production and intense work is expected to continue in this area. (Plant biotechnology, pp. 335–36, 337, 339–41)

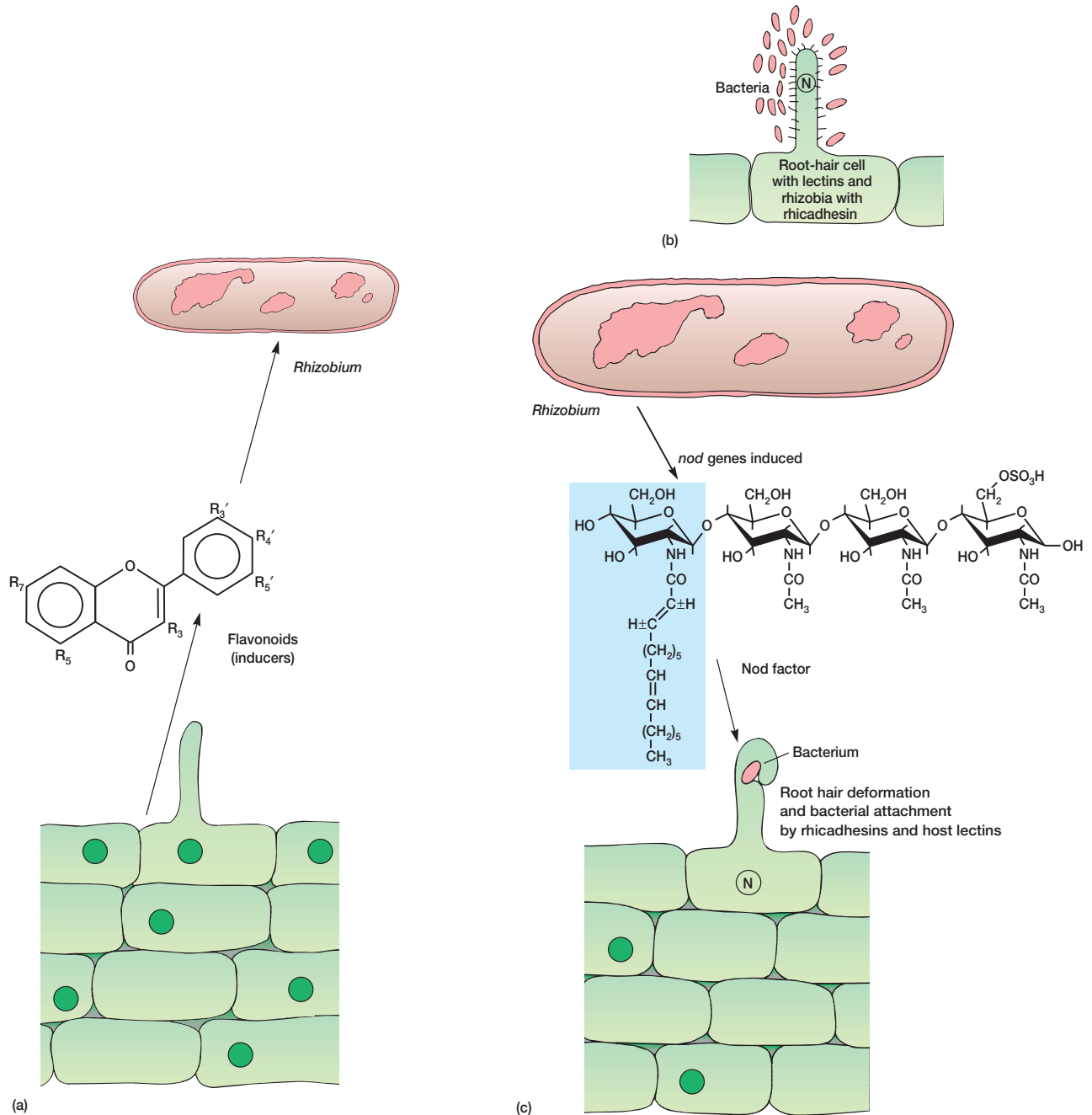
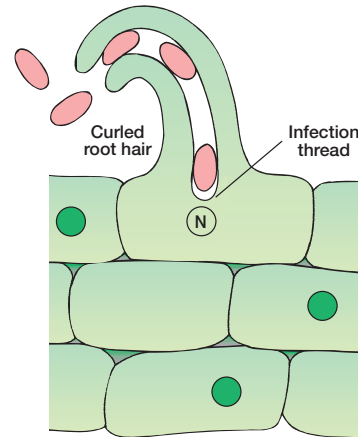


Figure 30.8 Root Nodule Formation by *Rhizobium*. Root nodule formation on legumes by *Rhizobium* is a complex process that produces the nitrogen-fixing symbiosis. (a) The plant root releases flavonoids that stimulate the production of various Nod metabolites by *Rhizobium*. There are many different Nod factors that control infection specificity. (b) Attachment of *Rhizobium* to root hairs involves specific bacterial proteins called rhicadhesins and host plant lectins that affect the pattern of attachment and *nod* gene expression. (c) Structure of a typical Nod factor that promotes root hair curling and plant cortical cell division. The bioactive portion (nonreducing *N*-fatty acyl glucosamine) is highlighted. These Nod factors enter root hairs and migrate to their nuclei. (d) A plant root hair covered with *Rhizobium* and undergoing curling. (e) Initiation of bacterial penetration into the root hair cell and infection thread growth coordinated by the plant nucleus "N." (f) A branched infection thread shown in an electron micrograph. (g) Cell-to-cell spread of *Rhizobium* through transcellular infection threads followed by release of rhizobia and infection of host cells. (h) Formation of bacteroids surrounded by plant-derived peribacteroid membranes and differentiation of bacteroids into nitrogen-fixing symbiosomes. The bacteria change morphologically and enlarge around 7 to 10 times in volume. The symbiosome contains the nitrogen-fixing bacteroid, a peribacteroid space, and the peribacteroid membrane. (i) Light micrograph of two nodules that develop by cell division ($\times 5$). This section is oriented to show the nodules in longitudinal axis and the root in cross section. (j) *Sinorhizobium meliloti* nitrogen-fixing nodules on roots of white sweet clover (*Melilotus alba*). (Continued)

Figure 30.8 (Continued)



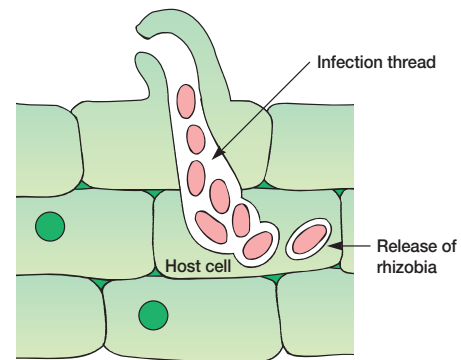
(d)



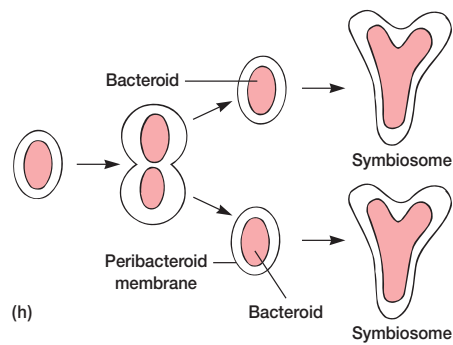
(e)



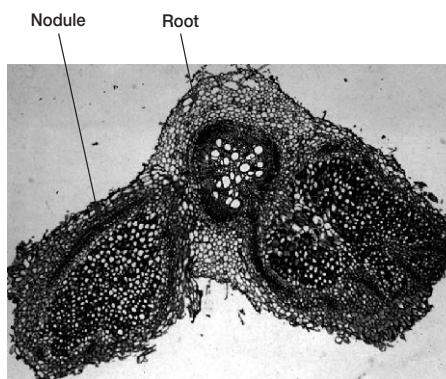
(f)



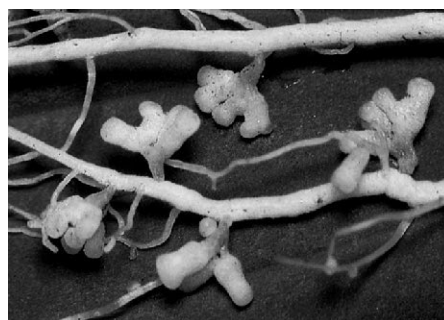
(g)



(h)



(i)



(j)

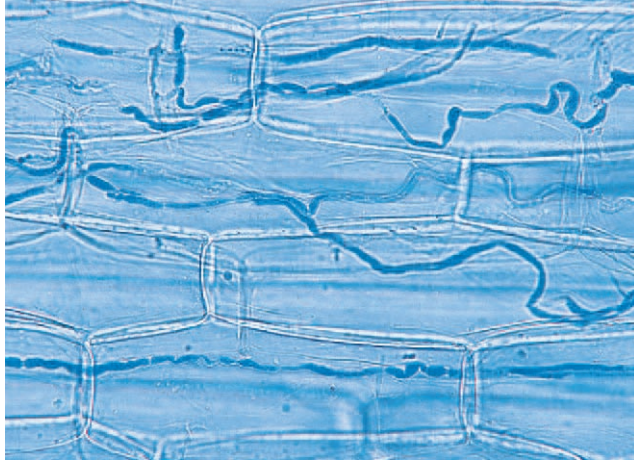


Figure 30.9 Fungal Endophytes. Fungi can invade the upper parts of some plants. A fungal endophyte growing inside the leaf sheath of a grass, tall fescue, is shown.



Figure 30.10 Parasitic Castration of Plants by Endophytic Fungi. Stroma of the fungus *Atkinsonella hypoxylon* infecting *Danthonia compressa* and causing abortion of the terminal spikelets.

Fungal and Bacterial Endophytes of Plants

Specialized fungi and bacteria can live within some plants as **endophytes** and may be beneficial. Specialized clavicipitaceous fungi form systemic fungal infections in which the endophyte grows between the plant's cortex cells (**figure 30.9**). Plants infected with these endophytes may be less susceptible to attack by various chewing insects due to the production of alkaloids, a form of "chemical defense." *Rhizobium leguminosarum* by *trifolii* can form a natural endophytic association with rice roots. This interaction, first observed in the Nile Delta, is supported by the rotation of clover with the rice. The association promotes rice root and shoot growth, resulting in increased rice grain yield at maturity. The rice/*Rhizobium*/clover association provides approximately one-quarter of the nitrogen needed by the rice crop.

Not all such relationships are mutualistic (beneficial to both partners). Some are parasitic. Parasitic fungal endophytes can actually reduce the genetic variability of the plant by sterilizing their host (**figure 30.10**). This "parasitic castration of plants" by systemic fungi, which promotes increased fungal spread in a less variable plant community, is suggested to be of major importance in the co-evolution of plants and fungi.

Endophytic bacteria have been discovered in sugar cane, cotton, pears, and potatoes. Some are plant pathogens that can survive for extended periods in a quiescent state. The majority have no known positive or deleterious effect on plant growth or development. The use of these bacteria as microbial delivery systems in agriculture is a current topic in agricultural biotechnology. It also has been possible to establish *Azorhizobium*, a root and stem nodulating bacterium for *Sesbana rostrata*, in the lateral roots of wheat plants, leading to a possible increased plant dry weight and nitrogen content.

1. Define and relate primordia and infection thread in terms of nodule formation.
2. What are possible effects of endophytic fungi on plants?
3. Give examples of genes that operate in microbe-plant and also in microbe-animal associations.
4. Describe the *nin* gene and its role in *Rhizobium* infection.
5. What are the basic steps in establishing the *Rhizobium*-legume root symbiosis?

Mycorrhizae

Mycorrhizae are fungus-root associations, first discovered by Albert Bernhard Frank in 1885 (**Box 30.1**). The term "mycorrhizae" comes from the Greek words meaning fungus and roots. These microorganisms contribute to plant functioning in natural environments, agriculture, and reclamation. The roots of about 95% of all kinds of vascular plants are normally involved in symbiotic associations with mycorrhizae.

Five mycorrhizal associations have been described (**figure 30.11**). These include both nonseptate and septate fungi. As noted in figure 30.11, there are endophytic arbuscular mycorrhizae (AM) that form arbuscules and sometimes vesicles (figure 30.11a), septate types associated with orchids (figure 30.11b), and those that form endomycorrhizal relationships with ericoid plants such as blueberries (figure 30.11c). In the endophytic mycorrhizae shown in figure 30.11a–c, the fungus penetrates the plant cells where it forms characteristic structures, including arbuscules and coils. Vesicles are not consistently observed. In addition, ectendomycorrhizae (figure 30.11d) are formed by basidiomycetes (*see p. 561*). These have sheaths and intracellular coils.

Box 30.1

Mycorrhizae and the Evolution of Vascular Plants

Fossil evidence shows that endomycorrhizal symbioses were as frequent in vascular plants during the Devonian period, some 387 to 408 million years ago, as they are today. As a result some botanists have suggested that the evolution of this type of association may have been a critical step in allowing colonization of the land by plants. During this period soils were poorly developed, and as a result

mycorrhizal fungi were probably significant in aiding the uptake of phosphorus and other nutrients. Even during current times those plants that start to colonize extremely nutrient-poor soils survive much better if they have endomycorrhizae. Thus it may have been a symbiotic association of plants and fungi that initially colonized the land and led to our modern vascular plants.

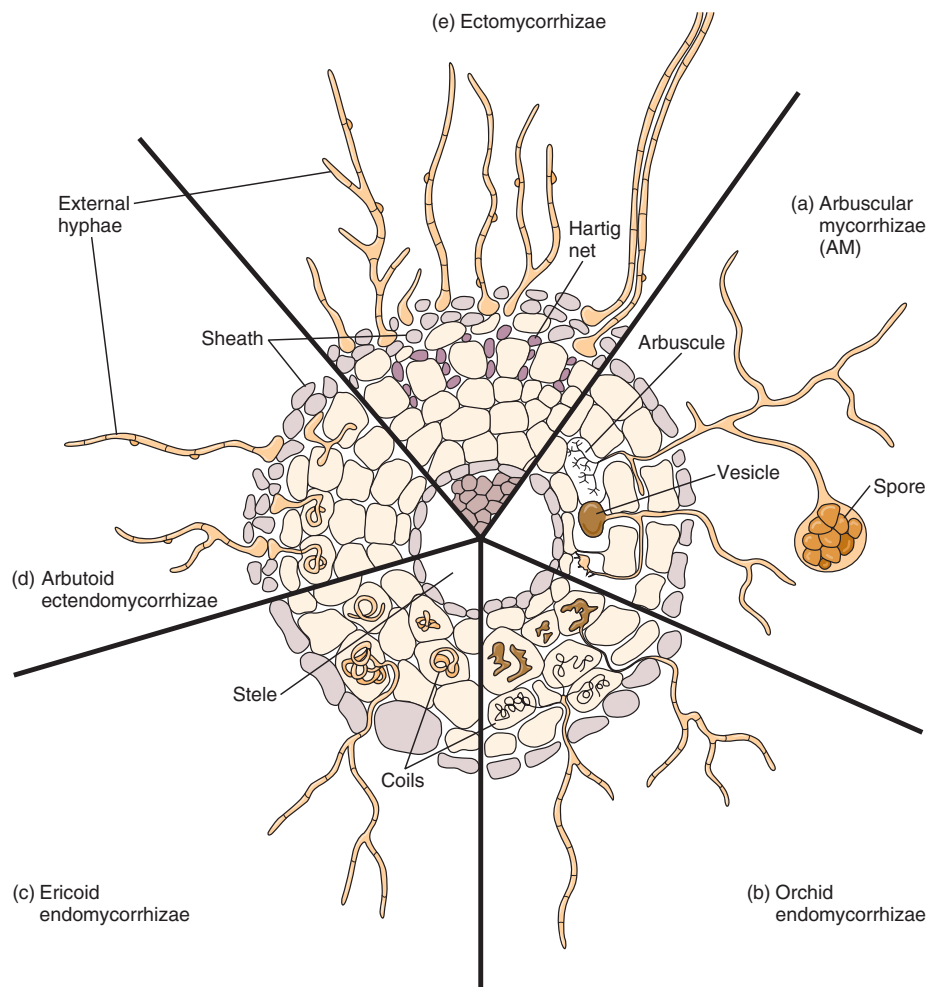


Figure 30.11 Mycorrhizae. Fungi can establish mutually beneficial relationships with plant roots, called mycorrhizae. Root cross sections illustrate different mycorrhizal relationships. See text for details.

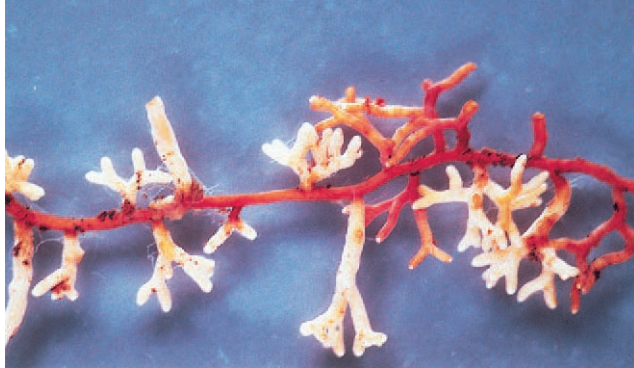


Figure 30.12 Ectomycorrhizae as Found on Roots of a Pine Tree. Typical irregular branching of the white, smooth mycorrhizae is evident.

Finally, the **ectomycorrhizae** (figure 30.11e) form a sheath, and the fungus grows between the plant cells, producing the “Hartig net.” Such ectomycorrhizae, including *Cennococcum*, *Pisolithus* and *Amanita*, form irregular structures that are easy to recognize (figure 30.12).

Bacteria also have relations with the mycorrhizal fungi. As the external hyphal network radiates out into the soil, a **mycorrhizosphere** is formed due to the flow of carbon from the plant into the mycorrhizal hyphal network and then into the surrounding soil. In addition, “mycorrhization helper bacteria” (MHB) can play a role in the development of the mycorrhizal relationships with the ectomycorrhizal fungus. Bacterial symbionts also are found in the cytoplasm of AM fungi, as shown in **figure 30.13**. By using PCR, such bacterial-like organisms (BLOs) appear to be related to *Burkholderia cepacia*. What is the possible function of these symbionts? It has been suggested that these “trapped” bacteria contribute to the nitrogen metabolism of the plant-fungal complex by assisting with the synthesis of essential amino acids.

More than 5,000 species of fungi, predominantly basidiomycetes, are involved in ectomycorrhizal symbiotic relationships. Their extensive mycelia extend far out into the soil and greatly aid the transfer of nutrients to the plant. One of the more important ectomycorrhizal fungi is *Pisolithus tinctorius*. This fungus can be grown in mass culture, with small Styrofoam beads acting as a physical support. The fungal inoculum is then mixed with rooting soil, resulting in improved plant establishment and growth.

Endomycorrhizae (figure 30.14) are of particular interest, as it has not been possible to grow these fungi, usually members of the zygomycetes, without the plant. In this association the fungal hyphae penetrate the outer cortical cells of the plant root, where they grow intracellularly and form coils, swellings, or minute branches. Endotrophic mycorrhizae are found in wheat, corn, beans, tomatoes, apples, oranges, and many other commercial crops, as well as most pasture and rangeland grasses. A characteristic intracellular structure is the **arbuscule**, and thus endomycorrhizae often are called **arbuscular mycorrhizal** or **AM fungi**. Recent studies show that plant flavonoids may stimulate spore germination, and this could lead to the development of plant-free cultures of these mycorrhizae.

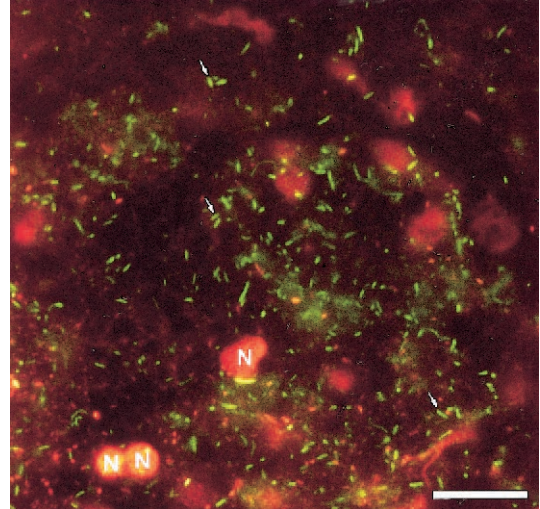


Figure 30.13 Mycorrhization Helper Bacteria. Stained bacterial endosymbionts in unfixed spores of the AM fungus *Gigaspora margarita*. Living bacteria fluoresce bright yellow-green; lipids and fungal nuclei (N) appear as diffuse masses. Bar = 7 μm .

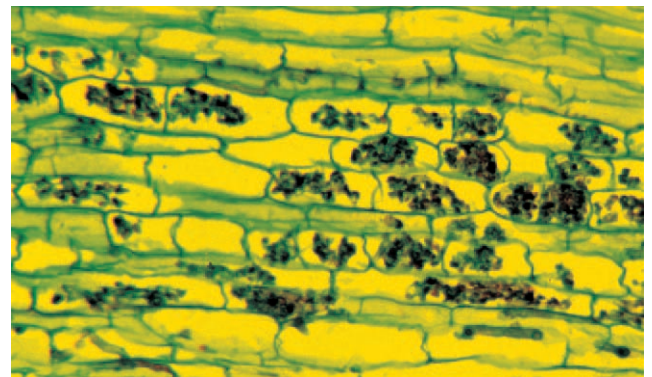


Figure 30.14 Endomycorrhizae. Endomycorrhizae, or arbuscular mycorrhizae, form characteristic structures within roots. These can be observed with a microscope after the roots are cleared and stained. The arbuscules of *Gigaspora margarita* can be seen inside the root cortex cells of cotton.

Depending on the environment of the plant, mycorrhizae can increase a plant's competitiveness. In wet environments they increase the availability of nutrients, especially phosphorus. In arid environments, where nutrients do not limit plant functioning to the same degree, the mycorrhizae aid in water uptake, allowing increased transpiration rates in comparison with nonmycorrhizal plants. These benefits have distinct energy costs for the plant in the form of photosynthate required to support the plant's “mycorrhizal habit.” Under certain conditions the plant is apparently willing to trade photosynthate—produced with the increased water acquisition—for water.

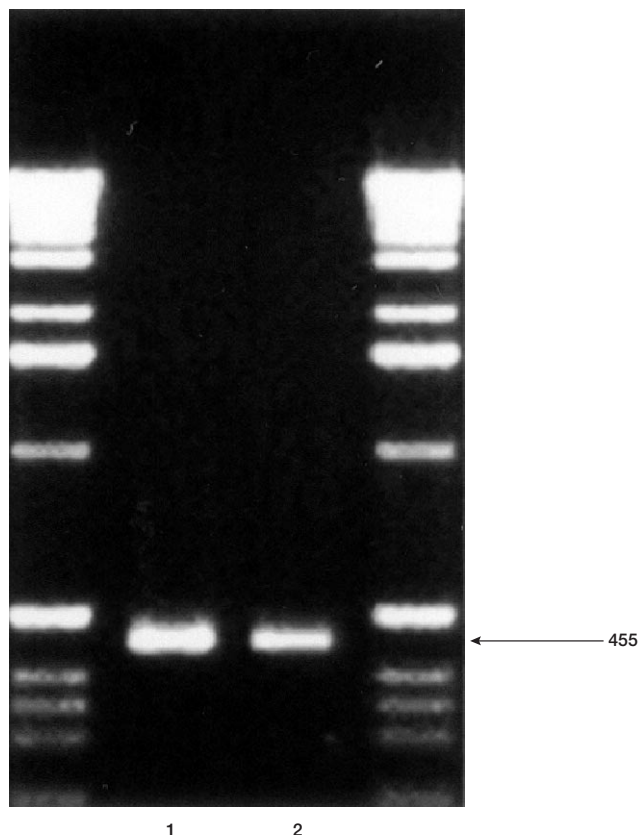


Figure 30.15 Use of the Polymerase Chain Reaction (PCR) to Detect a Mycorrhizal Fungus. Use of PCR to detect *Glomus intraradices* as spores (1) and after growth of the fungus in a root (2). In this example, a nested PCR was used. The second amplification is illustrated with the mycorrhizal fungal PCR product having a 455 base pair size. Molecular size markers are noted in the outer lanes.

Mycorrhizal and endophytic fungal communities have been found to influence the development of plant communities. The mycorrhizal fungi also can make it possible to share resources such as carbon minerals and water between different plants.

When studying mycorrhizal and endophytic relationships, molecular techniques are now used to identify spores and also mycorrhizal fungi within plant roots, as shown in **figure 30.15**. By the use of PCR (*see pp. 326–27*) with specific primers, it is possible to identify the presence of specific mycorrhizal fungi in association with a given plant after extraction of the rDNA from the root!

Actinorhizae

Actinomycete associations with plant roots, called **actinorhizae** or actinorhizal relationships, also occur (**figure 30.16**). These are formed by the association of *Frankia* strains (*see pp. 548–49*)



Figure 30.16 Actinorhizae. *Frankia*-induced actinorhizal nodules in *Ceanothus* (Buckbrush).

with eight nonleguminous plant families (**table 30.4**). They fix nitrogen and are important, particularly in trees and shrubs. As examples, these associations occur in areas where Douglas fir forests have been clear-cut, and in bog and heath environments where bayberries and alders are dominant. The nodules of some actinorhizae are marble sized (**figure 30.17**). Not impressed? Some plants (*Alnus*, *Ceanothus*) have nodules as large as baseballs. The nodules of *Casuarina* approach soccer ball size!

Members of the genus *Frankia* are slow-growing, and, until 1978, it was impossible to culture the organisms apart from the plant. Since then, this actinomycete has been grown on specialized media supplemented with metabolic intermediates such as pyruvate. Major advances in understanding the physiology, genetics, and molecular biology of these microorganisms are now taking place.

As in all plant-microbe associations, the actinorhizal relationship costs the plant energy. However, the plant does benefit and is better able to compete in nature. This association provides a unique opportunity for microbial management to improve plant growth processes.

Other associations of nitrogen-fixing microorganisms with plants also occur. A particularly interesting association is caused by **stem-nodulating rhizobia**, found primarily in tropical legumes (**figure 30.18**). These nodules form at the base of adventitious roots branching out of the stem just above the soil surface and, because they contain oxygen-producing photosynthetic tissues, they have unique mechanisms to protect the oxygen-sensitive nitrogen fixation enzymes. Another microorganism that forms such root and stem nodules is *Azorhizobium caulinodans*, which forms nodules on the tropical legume *Sesbania rostrata*.

[Nitrogen fixation: biochemical aspects \(pp. 212–14\); Nitrogen cycle \(pp. 615–16\)](#)

Recently, it has been shown that some of these stem-nodulating rhizobia are photosynthetic. Thus they can obtain their energy not only from the plant organic compounds, but also from the light!

Table 30.4 Nonleguminous Nodule-Bearing Plants with *Frankia* Symbioses^a

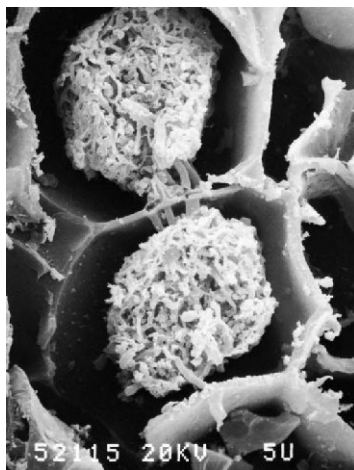
Family	Genus	<i>Frankia</i> Isolated?	Isolated Strains Infective?
<i>Casuarinaceae</i>	<i>Allocasuarina</i>	+	+
	<i>Casuarina</i>	+	+
	<i>Ceuthostoma</i>	—	—
	<i>Gymnostoma</i>	+	+
<i>Coriariaceae</i>	<i>Coriaria</i>	+	+
<i>Datisceae</i>	<i>Datisca</i>	+	—
<i>Betulaceae</i>	<i>Alnus</i>	+	+
<i>Myricaceae</i>	<i>Comptonia</i>	+	+
	<i>Myrica</i>	+	+
<i>Elaeagnaceae</i>	<i>Elaeagnus</i>	+	+
	<i>Hippophae</i>	+	+
	<i>Shepherdia</i>	+	+
<i>Rhamnaceae</i>	<i>Adolphia</i>	—	—
	<i>Ceanothus</i>	+	—
	<i>Colletia</i>	+	+
	<i>Discaria</i>	+	+
	<i>Kentrothamnus</i>	—	—
	<i>Retanilla</i>	+	+
	<i>Talguenea</i>	+	+
	<i>Trevoa</i>	+	+
	<i>Cercocarpus</i>	+	—
<i>Rosaceae</i>	<i>Chaemabatia</i>	—	—
	<i>Cowania</i>	+	—
	<i>Dryas</i>	—	—
	<i>Purshia</i>	+	—

Source: Data from Dr. D. Baker, MDS Panlabs and Dr. J. Dawson, University of Illinois. Personal communications.

^a*Frankia* isolation from nodules and ability of these isolated strains to initiate nodulation are also noted.



(a)



(b)

Figure 30.17 Actinorhizae. Nitrogen-fixing actinomycetes can form associations with woody plants. Actinorhizal nodule development allows symbiotic nitrogen fixation, critical for plant development on infertile sites. (a) Light micrograph of actinorhizae around *Comptonia* roots. (b) SEM view of two infected cortical cells of *Casuarina*. Note the hyphae from the actinorhizae penetrating the host's cell walls.

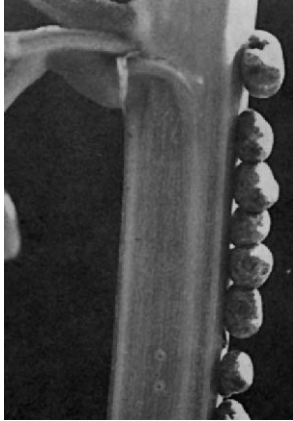


Figure 30.18 Stem-Nodulating Rhizobia. Nitrogen-fixing microorganisms also can form nodules on stems of some tropical legumes. Nodules formed on the stem of a tropical legume by a stem-nodulating *Rhizobium*.



Figure 30.19 Agrobacterium. *Agrobacterium*-caused tumor on a *Kalanchoe* sp. plant.

1. Describe the major contributions of mycorrhizae and actinorhizae to plant functioning.
2. Discuss the major differences between endomycorrhizae and ectomycorrhizae.
3. What is the major contribution of *Frankia* to plant functioning? Which types of plants are infected?
4. What are stem-nodulating rhizobia?

Agrobacterium

Another exciting plant-soil microorganism interaction is the *Agrobacterium* infection that produces tumorlike growths on plants (**figure 30.19**; see also *Box 14.2*). Gall formation involves *Agrobacterium* strains that contain the **Ti** (tumor-inducing) or **Ri** **plasmid**. Tumorigenesis is a natural form of genetic engineering, in which a portion of the Ti plasmid is excised, transferred from the bacterium to the plant host, and ligated into the host nuclear chromosome. The tumor is produced by an imbalance of phytohormones controlled by enzymes encoded by the introduced bacterial Ti DNA. Biotechnologists have modified this plasmid to allow the transfer of genetic characteristics such as herbicide resistance and bioluminescence to plants. With the ability to modify plant DNA using *Agrobacterium*-borne plasmids, rapid advances are being made in plant molecular biology. [Agrobacterium](#) (pp. 339–40, 492–93)

Tumor production involves binding of the bacterium to a plant, usually at a wound site, after plant phenolics are detected by the bacterium, which then expresses a tumor-inducing two-component system (**figure 30.20**). This oncogenic interaction is initiated by the transfer of tumor-DNA fragments (T-DNA) from the tumor-inducing (Ti) bacterial plasmid to the plant cell nucleus. One component controls the excision of the T-DNA, lead-

ing to the initiation of the tumor. The second component stimulates opine synthesis by the plant. The opine is used as a nutrient by the infecting *Agrobacterium*, but not as efficiently by other competing microorganisms. Thus plant metabolism is switched to produce substances that stimulate growth of the bacterium responsible for initiation of the tumor! This is a delicate interaction which has a genetic basis.

Fungi and Bacteria as Plant Pathogens

It should be noted that fungi can be devastating plant pathogens. Examples are the fungus *Puccinia graminis*, which causes wheat rust, and the fungus *Phytophthora infestans*, which was responsible for the Irish potato famine.

A wide variety of bacteria, including *Agrobacterium*, attack plants. Bacteria cause a bewildering array of spots, blights, wilts, rots, cankers, and galls, as shown in **table 30.5**. In addition, wall-less phytoplasmas are important pathogens of vegetable crops such as sweet potatoes.

Viruses

A wide range of viruses infect plants, as described in section 18.6, and are of worldwide importance in terms of plant disease and economic losses. These include TMV, the first virus to be characterized in 1892. A virus of particular interest in terms of plant-pathogen interactions is a hyphovirus that infects the fungus *Cryphanectria parasitica*, the cause of chestnut blight. Workers in Connecticut and West Virginia noted that if they infected the fungus with the hyphovirus, the rate and occurrence of infection was decreased. They are hoping to be able to treat trees with the less lethal virus strains and eventually transform the indigenous lethal strains of *Cryphanectria* into more benign fungi.

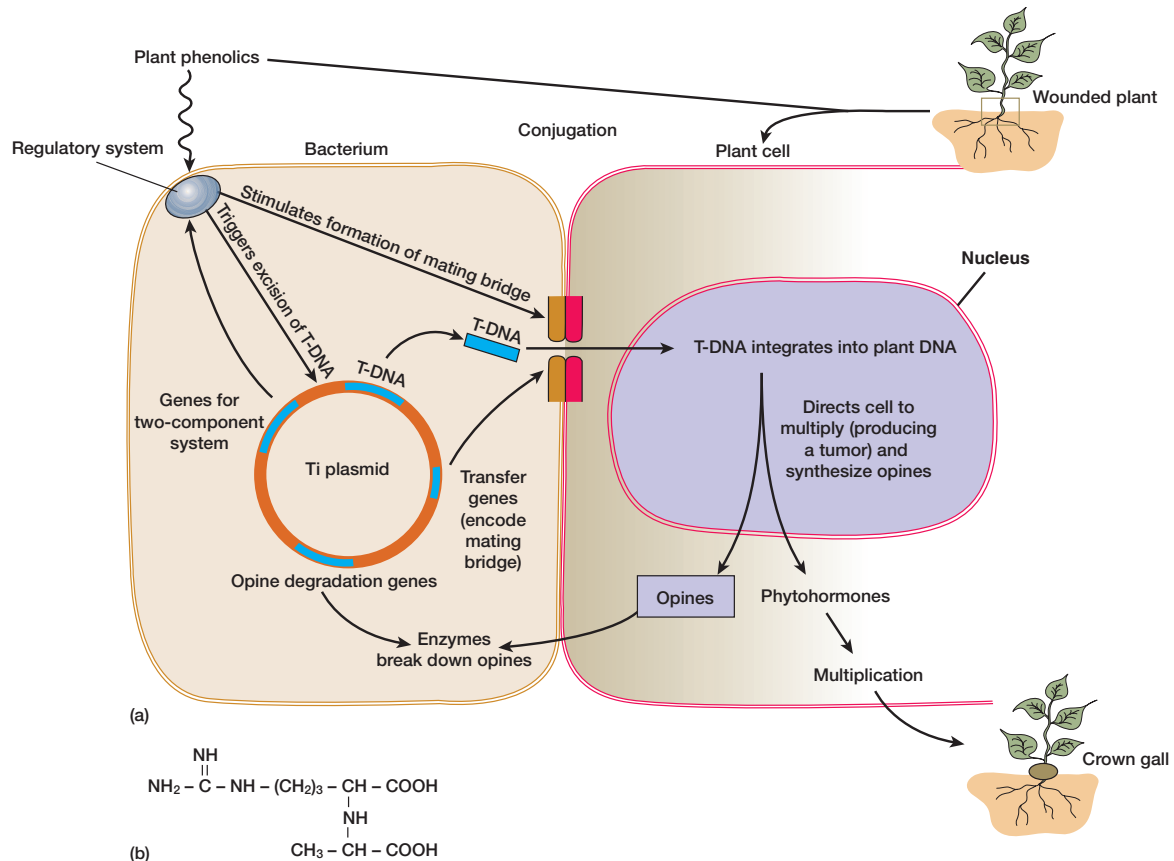


Figure 30.20 Functions of Genes Carried on the *Agrobacterium* Ti Plasmid. (a) Genes carried on the Ti plasmid of *Agrobacterium* control tumor formation by a two-component regulatory system that stimulates formation of the mating bridge and excision of the T-DNA. The T-DNA is moved by transfer genes, which lead to integration of the T-DNA into the plant nucleus. T-DNA encodes plant hormones that cause the plant cells to divide, producing the tumor. The tumor cells produce opines (shown in b) that can serve as a carbon source for the infecting *Agrobacterium*. This results in the formation of a crown gall on the stem of the wounded plant above the soil surface.

Tripartite and Tetrapartite Associations

An additional set of interactions occurs when the same plant develops relationships with two or three different types of microorganisms. These more complex interactions are important to a variety of plant types in both temperate and tropical agricultural systems. First described in 1896, these symbiotic associations involve the interaction of the plant-associated microorganisms with each other and the host plant. Several **tripartite associations** are known to occur: the plant plus (1) endomycorrhizae plus rhizobia, including *Rhizobium* and *Bradyrhizobium*; (2) endomycorrhizae and actinorhizae; and (3) ectomycorrhizae and actinorhizae. Nodulated and mycorrhizal plants are better suited for coping with nutrient-deficient environments. **Tetrapartite asso-**

ciations also occur. These consist of endomycorrhizae, ectomycorrhizae, *Frankia*, and the host plant. These complex associations, in spite of their additional energy costs, provide important benefits for the plant.

1. Discuss the nature and importance of the Ti plasmid.
2. What are the functions of the members of the two-component system in infection of a plant by *Agrobacterium*? What are the roles of phenolics and opines in this infection process?
3. Give some important fungal and bacterial plant pathogens.
4. How are plant pathologists attempting to control chestnut blight?
5. What are tripartite and tetrapartite associations?

Table 30.5 Major Plant Diseases Caused by Bacteria

Symptoms	Examples	Pathogen
Spots and blights	Wildfire (tobacco)	<i>Pseudomonas syringae</i> pv. ^a <i>tabaci</i>
	Haloblight (bean)	<i>P. syringae</i> pv. <i>phaseolica</i>
	Citrus blast	<i>P. syringae</i> pv. <i>syringae</i>
	Leaf spot (bean)	<i>P. syringae</i> pv. <i>syringae</i>
	Blight (rice)	<i>Xanthomonas campestris</i> pv. <i>oryzae</i>
	Blight (cereals)	<i>X. campestris</i> pv. <i>translucens</i>
	Spot (tomato, pepper)	<i>X. campestris</i> pv. <i>vesicatoria</i>
	Ring rot (potato)	<i>Clavibacter michiganensis</i> pv. <i>sepedonicum</i>
Vascular wilts	Wilt (tomato)	<i>C. michiganensis</i> pv. <i>michiganensis</i>
	Stewart's wilt (corn)	<i>Erwinia stewartii</i>
	Fire blight (apples)	<i>E. amylovora</i>
	Moko disease (banana)	<i>P. solanacearum</i>
	Black rot (crucifers)	<i>X. campestris</i> pv. <i>campestris</i>
	Soft rots (numerous)	<i>E. carotovora</i> pv. <i>carotovora</i>
Soft rots	Black leg (potato)	<i>E. carotovora</i> pv. <i>atroseptica</i>
	Pink eye (potato)	<i>P. marginalis</i>
	Sour skin (onion)	<i>P. cepacia</i>
Canker	Canker (stone fruit)	<i>P. syringae</i> pv. <i>syringae</i>
	Canker (citrus)	<i>X. campestris</i> pv. <i>citri</i>
	Crown galls (numerous)	<i>Agrobacterium tumefaciens</i>
Galls	Hairy root	<i>A. rhizogenes</i>
	Olive knot	<i>P. syringae</i> pv. <i>savastanoi</i>

Source: From J. W. Lengler, G. Drews, H. G. Schlegel. 1999. *Biology of the prokaryotes*, Blackwell Science, Malden, Mass., table 34.4.

^ap.v., pathover, a variety of microorganisms with phytopathogenic properties.

30.5 Soils, Plants, and Nutrients

Soil organic matter helps retain nutrients, maintain soil structure, and hold water for plant use. This important resource is subject to gains and losses, depending on changes in environmental conditions and agricultural management practices. Plowing and other similar disturbances expose the soil organic matter to more oxygen, leading to extensive microbiological degradation of organic matter. Irrigation causes periodic wetting and drying, which can also lead to increased degradation of soil organic matter, especially at higher temperatures.

Several approaches have been developed to overcome management practices that stimulate soil microorganisms and lead to increased degradation of soil organic matter. These approaches include “no till” or “minimum till” agriculture, in which surface soil is minimally disturbed and chemicals are used to control weed growth.

Composting is another, much older but still effective approach to maintaining and augmenting the organic matter content of a soil. In this process plant materials are allowed to decompose under moist, aerobic conditions, in which the readily usable plant fractions are rapidly decomposed. If the compost is too moist or too dry, the desired decomposition process will not occur. The compost pile reaches higher temperatures, allowing thermophilic microorganisms to participate actively in these processes. When the composting is completed, the residual

lignin and other more resistant plant materials will have been partially transformed to humus. Such biologically stabilized compost, when added to soil, increases the soil organic matter content and does not stimulate the soil microorganisms. [Thermophile characteristics](#) (pp. 125–26)

Throughout the world, soils are increasingly being impacted by mineral nitrogen releases resulting from human activities. This nitrogen is derived from two major sources: (1) agricultural fertilizers containing chemically synthesized nitrogen, and (2) fossil fuel combustion. It is interesting to compare the global distribution of releases from these two sources across the world (**figure 30.21**). North America, Europe, India, and eastern China are major sources of nitrogen from agriculture, whereas fossil fuel-based releases occur especially in eastern North America, Europe, eastern China, and Japan.

These nitrogen releases have had a wide variety of effects. Soils have a limited ability to tie up added nitrogen compounds in organic matter; they are limited by the amount of carbon that can be incorporated into microbial biomass and can reach the **nitrogen saturation point**. Beyond this point, added nitrogen will not be incorporated into organic matter and remains in a mobile form. The major types of nitrogen fertilizers used in agriculture are liquid ammonia and ammonium nitrate. Ammonium ion usually is added because it will be attracted to the negatively charged clays in a soil and be retained on the negative clay surfaces until used

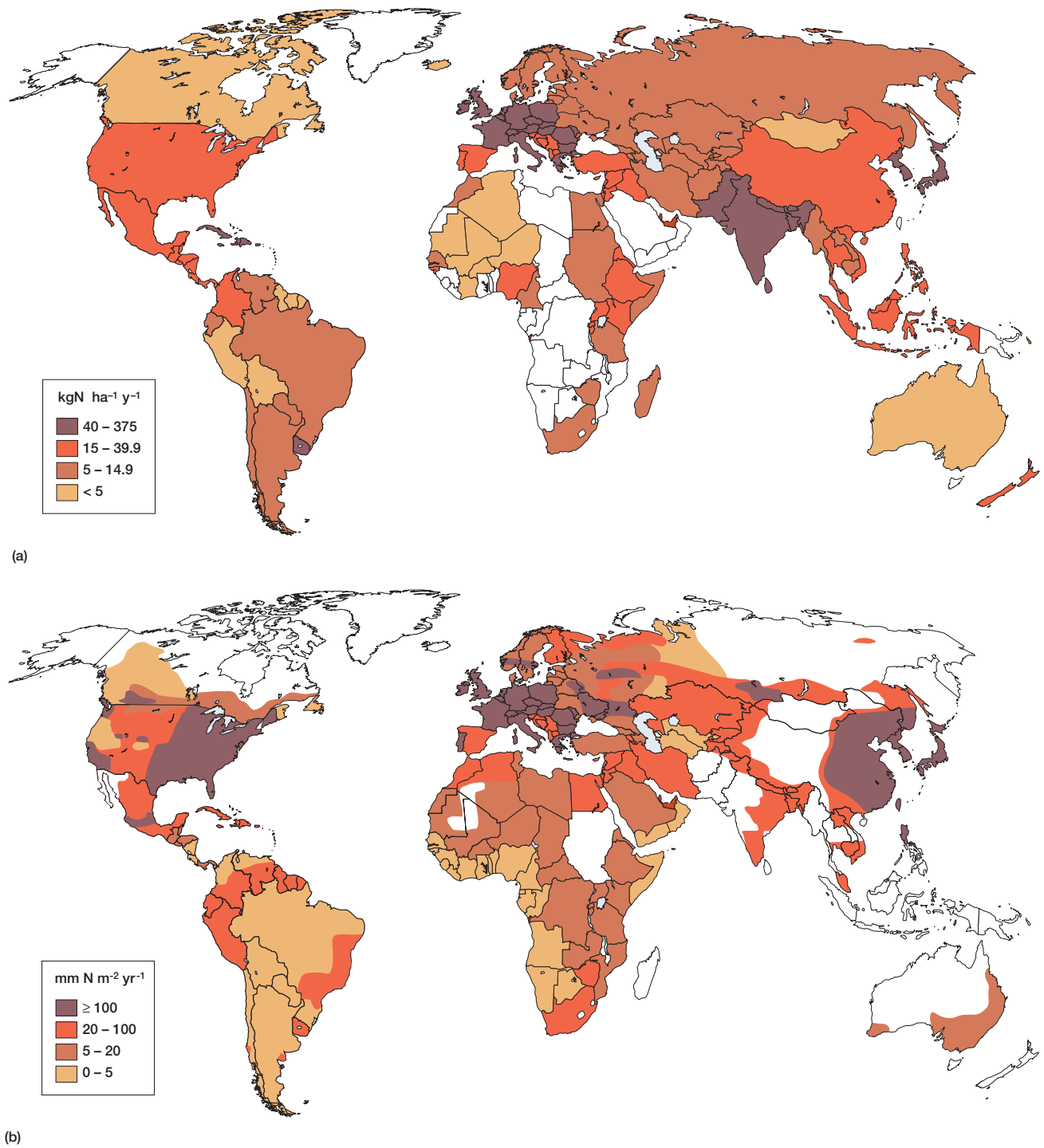
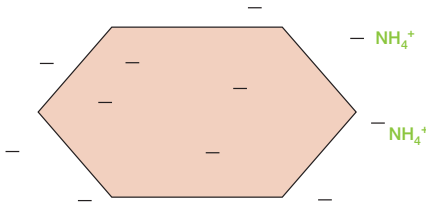


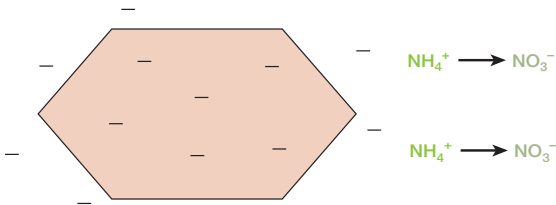
Figure 30.21 Global Releases of Nitrogen to the Environment from Agriculture and Fossil Fuels.

Releases of nitrogen to the environment from agriculture and animal/human wastes for 1987 to 1990 have a somewhat different global distribution than the releases from fossil fuel burning. **(a)** Releases of agricultural and animal wastes from the Caribbean, Europe, India/Pakistan, and Japan/Korea are intense. **(b)** In contrast, fossil fuel combustion releases, expressed as $\text{mmol N m}^{-2} \text{yr}^{-1}$, are primarily from developed/developing areas of the world with high population densities. Adapted from: Nixon, S. W. 1995. *Coastal marine eutrophication: a definition, social causes, and future concerns*. *Ophelia* 41:199–219.

(a) Fertilizer-derived ammonium ions enter the soil and are attracted to negatively charged clays in the plant root zone.



(b) Nitrification occurs in aerobic zones, transforming ammonium ions to the anionic nitrate.



(c) Nitrate, repelled from the negatively charged clays, moves with the soil water out of the plant root zone.

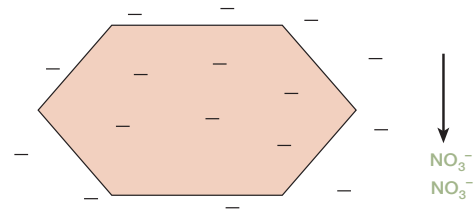


Figure 30.22 Fertilizer Mobility and Soil Clays. The role of negatively charged clays in (a) retaining ammonium ion-containing fertilizer in a soil, and (b) the release of nitrate, formed by nitrification of the ammonium ions, to the soil water and the environment. (c) Nitrate, being anionic or negatively charged, is not retained by the negatively charged soil clays.

as a nutrient by the plants (figure 30.22). However, the nitrifier populations in a soil (see pp. 615–16) can oxidize the ammonium ion to nitrite and nitrate, and these anions can be leached from the plant environment and enter surface waters and groundwaters.

When nitrogen fertilizers are used, an inevitable consequence has been higher nitrate levels in waters, which can contribute to infant respiratory problems, and possibly to the production of nitrites and the formation of nitrosamine carcinogens. In addition, plants grown in high nitrate-concentration soils may accumulate nitrate to a level that is harmful for animals. Cereal grains, many weeds, and grass hay contain high nitrate levels when grown in such soils.

Nitrogen fertilizers also can affect the microbial community structure and function. The use of nitrogen fertilizers has led to decreases in filamentous fungal development in a wide variety of soils. The loss of fungi affects the soil structure because fungi link soil particles to stabilize the soil peds. With a weakened and decreased fungal community, mycorrhizal-dependent plants become more susceptible to stresses such as drought and toxic metals.

As a final note, excessive fertilizer use can have global-level impacts when nitrogen enters coastal and marine waters (see chapter 29). The chemical synthesis of mineral fertilizers, a heritage of the Haber Process developed early in the twentieth century, has had many unexpected global-level effects.

Phosphorus in fertilizers also is critical. The binding of this anionic fertilizer component to soils is dependent on the cation exchange capacity (CEC) and soil pH. As the soil phosphorus sorption capacity is reached, the excess, together with phosphorus that moves to waters with soil erosion can stimulate the growth of freshwater organisms, particularly cyanobacteria, in the process of eutrophication. The cyanobacteria are then able to fix more nitrogen, because in most freshwater systems, phosphorus is the critical limiting element.

1. What soil management processes can lead to increases in degradation of soil organic matter?
2. Why might composted materials have different effects on soil microorganisms and organic matter than additions of fresh plant materials?
3. What are possible effects of nitrogen-containing fertilizers on microbial communities?
4. Most nitrogen fertilizer is added as ammonium ion. Why is this preferred over nitrate?
5. Why is nitrate of concern when it reaches rivers, lakes, and groundwaters?
6. Why might enrichment of fresh waters with phosphorus be even more critical than nitrogen enrichment?

30.6 Soils, Plants, and the Atmosphere

Soil microorganisms have interesting interactions with the atmosphere, another critical part of aerobiology. Ice-nucleating bacteria from decomposing litter can increase precipitation. Specific strains of *Pseudomonas syringae* synthesize proteins that can serve as nucleation centers for snow formation in cloud chambers. Such ice-nucleating soil microorganisms may significantly influence weather on a global scale. “Ice-minus” strains of *P. syringae*, developed by recombinant technology, can be sprayed on frost-sensitive crops such as strawberries. In the presence of the “ice-minus” bacteria, water will not freeze on the leaves as readily, making it possible to save frost-sensitive crops from destruction. The release of such genetically modified microorganisms (GEMs) to the environment is of continuing concern (see pp. 341–42).

Soil microorganisms also influence the atmosphere by degrading airborne pollutants such as methane, hydrogen, CO, benzene, trichloroethylene (TCE), and formaldehyde. They can substantially improve the air in closed buildings (Box 30.2). Although soil microorganisms cannot completely eliminate airborne pollutants, they can decrease these to equilibrium levels of approximately 1 to 2 ppm.

Soil microorganisms, through their activities in biogeochemical cycling (see section 28.3), also can have major effects on

Box 30.2

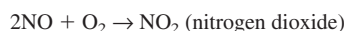
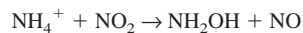
Keeping Inside Air Fresh with Soil Microorganisms

A major problem in the development of more energy-efficient homes and office buildings is the potential effect of such closed environments on human health. With many people spending much of their lives in such enclosed environments, the “sick building syndrome” is an increasing concern. While saving energy, these “sick buildings” have higher levels of many volatile compounds, including benzene, trichloroethylene (TCE), formaldehyde, phenolics, and solvents. These are released from rugs, furniture, plastic flooring, paints, and office machines such as photocopiers and printers. An important but

still largely unappreciated means of improving the air in such “sick buildings” is through plants and their associated soil microorganisms. Plants not only produce oxygen, but the soil microorganisms degrade many airborne pollutants. It is recommended that one plant be used per 100 square feet of living area. As noted by B. C. Wolverton, “The ultimate solution to the indoor air pollution problem must involve plants, the plant soils and their associated microorganisms.” Soil microorganisms, especially in association with plants, can help keep air in closed environments fresher and more healthful (plants are also nice to look at).

global fluxes of a variety of gases. These gases can be considered as those that are “relatively stable” and those that are “reactive gases.” Relatively stable gases that are influenced by microbial activities include carbon dioxide, nitrous oxide, nitric oxide, and methane. Microorganisms also contribute to the flow of reactive gases such as ammonia, hydrogen sulfide, and dimethylsulfide, as major examples. These reactive gases tend to be produced in more waterlogged environments.

Addition of nitrogen-containing fertilizers also affects atmospheric gas exchange processes in a soil. Nitrogen additions stimulate the production of the nitrification intermediates NO and N₂O, which are critical as **greenhouse gases**. NO also appears to be required for *Nitrosomonas eutropha* to carry out nitrification. The oxidation of NH₄⁺ involves the formation of hydroxylamine (NH₂OH) and NO as intermediates. This NO reacts with oxygen to give NO₂, which then can repeat the process in the following reaction:



In this sequence molecular oxygen does not react with NH₄⁺ but with NO. If NO is absent the reaction will not proceed. **Nitrification** (p. 193)

Atmospheric gases such as carbon dioxide, nitrous oxide, nitric oxide, and methane, that decrease the loss of radiation from the earth and may produce atmospheric warming, are called greenhouse gases. The production and consumption of these greenhouse gases can be influenced by a range of human activity-related factors. These include plant fertilization and automobile use, conversion of soils to agricultural use, and the use of landfills, all of which can cause increases in mineral nitrogen levels.

Methane is a greenhouse gas of increasing concern that can be derived from a variety of sources. These include ruminants, rice paddies, and landfills. Landfills, especially, can release methane to the atmosphere over longer terms (decades, centuries). Another interesting source of atmospheric methane is microorganisms that inhabit the gut of wood-eating termites. As noted in **Box 30.3**, with increased deforestation and the accumulation of plant residues, populations of termites (and termite gut-inhabiting methanogenic microorganisms) are increasing.

Methane levels are influenced by microorganisms and their functioning in the environment. Well-drained, aerobic soils are capable of methane oxidation by **methanotrophs** (see pp. 502–3), whereas in water-saturated sites of soils and in bogs and wetlands, methane may be produced faster than it can be used by methanotrophs. Based on analyses of gas bubbles in glacier ice cores, the levels of methane in the atmosphere remained essentially constant until about 400 years ago. Since then the methane level has increased 2.5-fold to the present level of 1.7 parts per million (ppm) by volume. Considering these trends, there is a worldwide interest in understanding the factors that control methane synthesis and use by microorganisms.

The processes of methane synthesis and use occur on a variety of scales in upland soils and in wetlands, as shown in **figure 30.23**. In nominally aerobic upland soils, there may be anaerobic hot spots (water-saturated local areas) where methane is produced. If these areas are surrounded by aerobic soils, methanotrophs degrade most of the methane before it can be released to the atmosphere. In more waterlogged areas such as lowland soils, methanogenesis can proceed and there is less opportunity for methanotrophs to function. In spite of this, most of the methane will be degraded. Aquatic plants transport oxygen to their rhizospheres and thus encourage methane oxidation in these localized aerobic hot spots. Methane oxidation also is sensitive to nitrogen-containing fertilizers and increases in atmospheric CO₂ levels. The balance between methane synthesis and degradation is very important. It is estimated that soils of all types provide 60% of the total atmospheric methane; wet areas, water-saturated hot spots, and rice paddies are particularly significant contributors.

1. What is a greenhouse gas? Give examples and discuss their possible importance.
2. Discuss the possible role of termites in the production of greenhouse gases.
3. What microbial processes occur in soils to both produce and degrade methane?
4. Describe factors that might lead to the formation of localized hot spots for the production and consumption of greenhouse gases.

Box 30.3

Soils, Termites, and Intestinal Microorganisms:
Understanding Atmospheric Methane Changes

Termites are important components of tropical ecosystems, where their use of cellulose plant materials allows rapid—sometimes too rapid—recycling of plant materials. Termites harbor significant populations of archaea that use products of cellulose digestion, including CO₂ and hydrogen, to produce methane.

Termites occur on 2/3 of the Earth's land surface, and, based on laboratory studies, 0.77% of the carbon ingested by termites can be released as methane. In tropical wet savannas and cultivated areas, termite popu-

lations are increasing rapidly. This increase is being accelerated by the destruction of tropical forests, which results in the accumulation of dead plant materials on the soil surface. This provides an ideal environment for the growth of termites. Termites are estimated to be contributing annually at least 1.5×10^{14} grams of methane, together with hydrogen and CO₂, to the atmosphere. This is believed to be contributing to measurable increases in the atmospheric methane level. Thus unseen termites and their associated gut microorganisms may be affecting global warming.

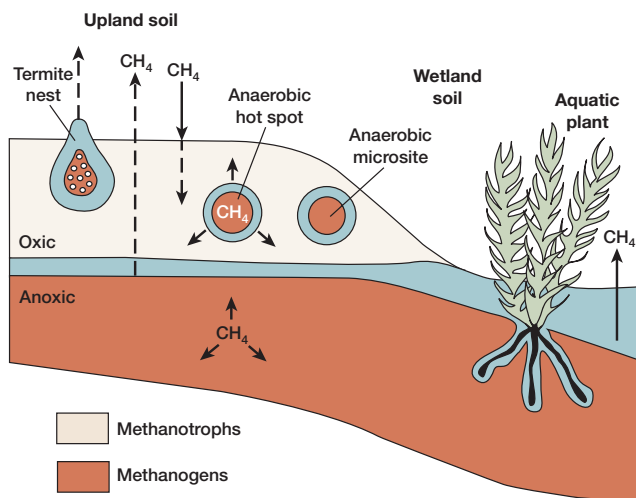


Figure 30.23 Methane Production and Use in Soils. Methane production and degradation can occur in closely located aerobic and anaerobic zones. In upland soils, methane synthesis can take place in localized anaerobic hot spots and termite nests. Much of this methane can be degraded in the surrounding aerobic soils. In wetland areas, methane production dominates in the waterlogged areas, and methane has a greater chance of being released to the atmosphere. If aquatic plants are present and translocate oxygen into anaerobic zones, these create localized aerobic hot spots in the rhizosphere for methane oxidation.

30.7 Microorganisms and Plant Decomposition

Plant growth is important in terrestrial ecosystems; equally important are plant death, decomposition, and recycling. Plants are constantly releasing leaves, branches, and other plant parts which enter the soil. These released soluble materials create a **residue-sphere**, an area between decaying plant material and the soil. This

has been described as a hotbed of microbial activity, which drives a wide range of microbiological processes. In anaerobic “hot spots” soluble plant materials support denitrification and possibly genetic exchange processes. After soluble materials have leached from the plant, the remaining starch, cellulose, and proteins are degraded both aerobically and anaerobically. Lignin, a random aromatic cyclic polymer that contains no nitrogen, is the one exception: its degradation requires oxygen. The basidiomycetes, which function best aerobically, are the major producers of laccases and the phenoloxidase enzymes required for lignin degradation. Lignin decomposition also is limited by the physical nature of the material: healthy woody plants are saturated with sap, creating an aerobic, low oxygen flux environment similar to that found in waters (see chapter 29). In addition, high ethylene and CO₂ levels, as well as the presence of phenolic and terpenoid compounds, retard the growth of lignin-degrading fungi. Thus the first step in lignin decomposition is physical rot, which involves cavitation or the creation of an “air embolism” in vascular tissue that begins when sap ceases flowing through the structure. This provides sufficient oxygen within the plant tissue to allow lignin biodegradation.

Unusual gases are produced, particularly by fungi, in this process of woody plant decomposition; these include chloromethanes and cyanide, compounds that we normally associate with industrial pollution. Large amounts of CH₃Cl, an important greenhouse gas, are produced by many fungi, including the basidiomycetes *Phellinus* and *Inonotus*, which are members of the wood-rotting *Hymenochaetaceae*. The global input of CH₃Cl to the atmosphere from plant decomposition is thought to be 160,000 tons, 75% of which is derived from tropical and subtropical soils. It is estimated that 15 to 20% of the chlorine-catalyzed ozone destruction is due to naturally-produced chlorohydrocarbons. **Basidiomycetes (p. 561)**

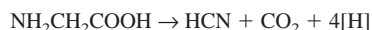
As noted in **table 30.6**, terrestrial environments, the oceans, and biomass burning are all important sources of atmospheric chloromethane. Large amounts also have been detected in some basidiomycetes. These include concentrations of 74 to 2,400 mg/kg in the basidia of some agarics and bracket fungi. It is of interest that the maximum permissible chlorophenol levels in soils

Table 30.6 Global Emissions of Chloromethane

Source	Inputs to Atmosphere (10 ⁵ tons/year)
Natural Sources	
Terrestrial processes	0–20
Ocean fluxes	3–20
Biomass burning	4–14
Anthropogenic Sources	0–3

Source: From R. Watling and D. B. Harper, 1998, *Mycol. Res.* 102(7):769–87.
*Estimated atmospheric inputs from natural and anthropogenic sources.

are only 1 to 10 mg/kg! Cyanide is another chemical of widespread concern produced by fungi, especially by basidiomycetes and ascomycetes. This cyanide may be derived from the S-methyl group of L- and D-methionine, or its production may be linked to methyl benzoate synthesis. The best studied cyanide-producing fungi are *Marasmius oreades* (the cause of fairy ring disease) and the snow mold basidiomycetes. Some bacteria also produce cyanide. Cyanide synthesis involves the oxidative decarboxylation of glycine, which is stimulated by methionine or other methyl-group donors in the following reaction:



Cyanide can inhibit respiration. It also can serve as a carbon and nitrogen source for microorganisms, including cyanogenic fungi such as *Marasmius* and *Pholiota*, and some actinomycetes. This illustrates the adaptability of microorganisms in the use of a nominally toxic metabolic product.

1. Why are fungi critical for the degradation of woody plants? How does the plant defend itself against these fungi?
2. What is the residuesphere?
3. What critical products of fungal growth on wood are usually considered only to be industrial pollutants?
4. How do microorganisms contribute to the cycling of cyanide?

30.8 The Subsurface Biosphere

An important habitat for microbial growth is the **subsurface biosphere**, a source of many questions concerning the origin of life on the Earth. The subsurface biosphere has been studied by examining outcrops, surface excavations, petroleum hydrocarbons, well corings, and materials from deep mine sites. Drillers have reached a depth of 6,100 meters (20,000 feet) at the Inigok well, North Slope, Alaska, and 9,150 meters (30,000 feet) at the Becha Rogers well in the Anadarko Basin, located in northern Texas and Oklahoma. In a South African gold mine, rocks from deeper than 3,350 meters (11,000 feet) below the surface have

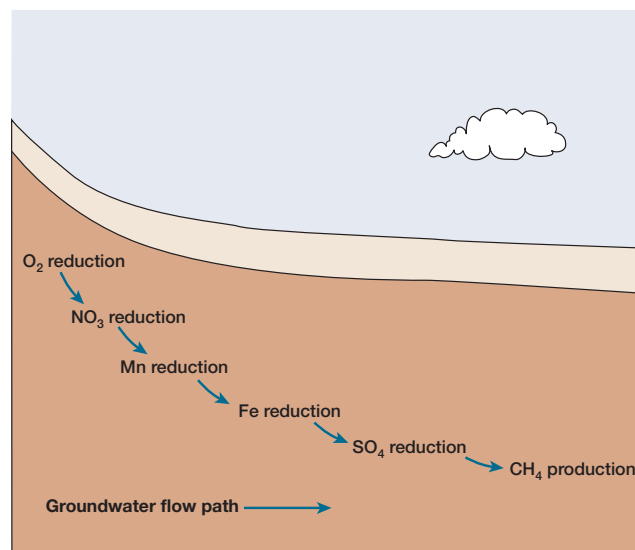


Figure 30.24 The Shallow Subsurface Biosphere. The shallow subsurface, as in a stable sediment, showing the distribution with depth of oxidants that can occur in an aerobic pristine aquifer. In aerobic sediments, the oxidants will be distributed with the most energetically favorable (oxygen) near the surface and the least energetically favorable at the lower zones of the geological structure. Source: Lovley, D. K., 1991. *Dissimilatory Fe (III) and Mn (IV) reduction. Microbiol. Rev.* 55:259–287.

been recovered by hand and examined in the search for new microorganisms. A long history of interest in the subsurface preceded these efforts, and there have been reports of microorganisms in rocks, particularly from rocks obtained in deep drilling programs. Much of this information has been discounted as being due to drilling contamination or water flows from surface areas. In recent years the use of improved techniques, including molecular approaches, has led to a greater acceptance of the idea of microorganisms occurring far below the Earth's surface, leading to a new field: deep subsurface microbiology.

Microbial processes take place in different subsurface regions, including (1) the shallow subsurface where water flowing from the surface moves below the plant root zone; (2) subsurface regions where organic matter, originating from the Earth's surface in times past, has been transformed by chemical and biological processes to yield coal (from land plants), kerogens (from marine and freshwater microorganisms), and oil and gas; and (3) zones where methane is being synthesized as a result of microbial activity.

In the shallow subsurface, surface waters often move through **aquifers**, porous geological structures below the plant root zone. As shown in **figure 30.24**, in a pristine system with an aerobic surface zone, the oxidants used in catabolism will be distributed from the most oxidized and energetically favorable (oxygen) near the surface to the least favorable (in which CO_2 is used in methanogenesis) in lower zones.

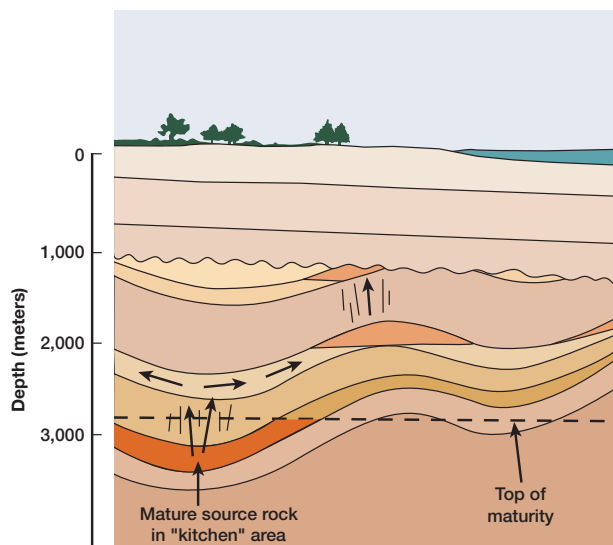


Figure 30.25 The Oil and Gas Region of the Subsurface. Organic matter, originating from the Earth's surface, is transformed to oil, gas and coal by chemical, thermal, and biological processes. Above the higher-temperature "kitchen" area, where chemical changes occur, microorganisms can contribute to the processing of these organic materials. Hydrocarbons migrate through porous strata and fractures, finally accumulating in porous, overlying geological structures. Lines indicate fractures, and arrows indicate hydrocarbon flows.

In subsurface regions where organic matter from the Earth's surface has been buried and processed by thermal and possibly biological processes, kerogen and coals break down to yield gas and oil, as noted in **figure 30.25**. After their generation, these mobile products, predominantly hydrocarbons, move upward into the more porous geological structures where microorganisms can be active. Chemical signature molecules from plant and microbial biomass are present in these petroleum hydrocarbons.

Below these zones lie vast regions where methane is present in geological structures (**figure 30.26**); this methane is continuously being released to the overlying strata. Based on studies of stable carbon isotopes, in the "biogenic" zone, methane has less ^{13}C isotope, indicating that it was derived from microorganisms using H_2 as an energy source and CO_2 as a carbon source and oxidant. Given a choice, microbes will tend to use the lighter of two isotopes—in this case ^{12}C over the ^{13}C isotope-containing CO_2 . In the underlying hotter abiogenic zone, in comparison, the methane is not depleted of the heavier carbon isotope, indicating that this is of chemical and thermal origin.

Microorganisms also appear to be growing in intermediate depth oil-bearing structures. Recent studies indicate that active procaryotic assemblages are present in high-temperature (60 to 90°C) oil reservoirs, including such genera as *Thermotoga*, *Ther-*

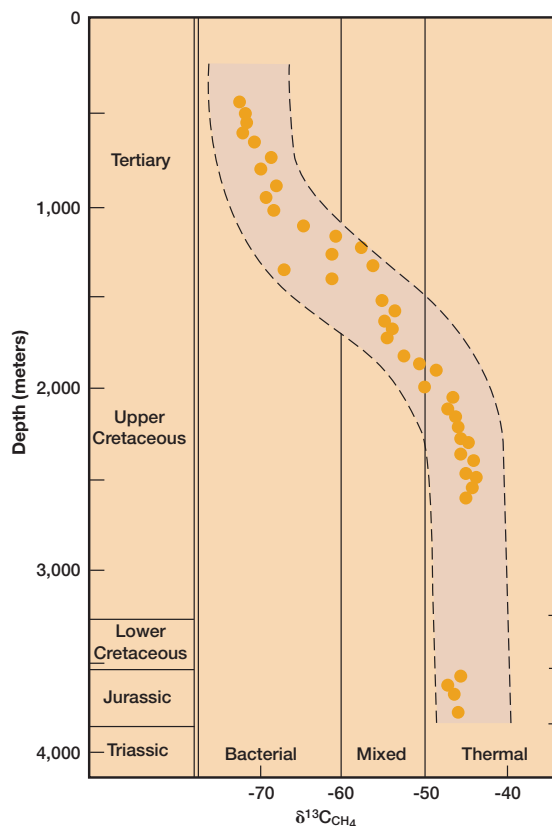


Figure 30.26 Methane Synthesis by Microorganisms in the Subsurface. The microbially mediated production of methane has been shown to occur in the subsurface by the use of stable isotope techniques. The decrease in the occurrence of the ^{13}C isotope of carbon indicates that the methane was produced by microorganisms down to a depth of 1,500 meters under the floor of the North Sea. Below a depth of 2,000 meters, the methane does not have a lower frequency of the ^{13}C isotope of carbon, indicating that it was formed by abiogenic processes. The $\delta^{13}\text{C}$ value gives an indication of the relative proportion of ^{13}C to ^{12}C in the sample. The more negative scale values signify a decreased presence of the heavier isotope in the upper zone at this location.

moanaerobacter, and *Thermococcus*. The archaean genera are dominated by methanogens. Thus microbial activities may be occurring above or in the "deep hot biosphere," a term suggested by Thomas Gold to describe this poorly understood region.

1. What types of microbial activities have been observed in the deep subsurface?
2. What happens in terms of microbiological processes when organic matter leaches from the surface into the subsurface?
3. Why are stable isotope analyses so important in studies of microbe-geological interactions?
4. What microbial genera have been observed in oil field materials?

30.9 Soil Microorganisms and Human Health

As with waters, humans are in constant contact with soils. This occurs directly as when children or adults play in the “dirt,” or even when leafy and root vegetables, covered with soil dust, are eaten. In most cases the contact with soil is harmless. However, soils do contain a wide variety of pathogenic organisms. What is needed is an entry point and favorable conditions within or on the human body. A wide variety of anaerobes, including *Clostridium*, are present in soils. Unless there is a deep puncture wound that will provide the anaerobic environment required for their growth, anaerobes are of little concern. However, puncture wounds, as occur in warfare and accidents, can lead to gangrene. This was an important cause of military fatalities prior to the advent of modern microbiology and antibiotics. Soils contain other pathogens. Organisms such as *Acanthamoeba*, which can be inhaled from dust, may cause primary amebic meningoencephalitis. When soils are used for surface disposal of human wastes without sewage treatment, the transmission of a wide variety of pathogens, including protozoa such as *Acanthamoeba* and *Cyclospora* can occur. The use of “night soil” as fertilizer has led to widespread contamination of fresh fruits and vegetables and has had international repercussions.

Soil and soil-related microorganisms also are of concern when they grow in buildings (figure 30.27). This increasingly common problem, often linked to the flooding of houses located in low-lying districts or to moisture accumulation in sink and bathroom areas (even in large and modern homes), has led to major health problems. This is particularly severe when water penetrates into house walls and insulation materials. The problem has reached the point that national newsmagazines have published articles, showing large new expensive homes, entitled “The Mold in Your Home May Be Deadly!” In a recent article it was noted that 50% of homes have mold problems, a major source of chronic sinus infections. These molds also have been related to increases in asthma rates. The major responsible fungi are *Stachybotrys chartarum*, *Eurotium herbariorum*, and *Aspergillus versicolor*. Fungal growth results in a black slime; when this fungal growth dries, a dry dusty layer remains and the spores can be dispersed into the air. These spores are particularly dangerous for infants, whose lungs are less developed. *Stachybotrys* infection can result in pulmonary hemosiderosis, which causes bleeding of the lungs and sometimes death. Rapid drying of water-damaged buildings is required to control this problem. Recently a bioassay for such “sick buildings” has been developed using boar spermatozoa! In this analysis the motility of the spermatozoa is inhibited by extracts from moist fungal-infested materials. Items of particular concern are moldy carpets as well as piles of old newspapers.

It has been reported that wallboards can contain baseline bioburdens of these organisms, which need only high humidity to trigger fungal growth. In addition, *Mycobacterium komossense* and gram-negative endotoxin-producing bacteria have been isolated. There are limited alternatives for controlling and removing such dangerous fungi; the most important are the removal/disinfection of moldy materials and keeping a house dry.

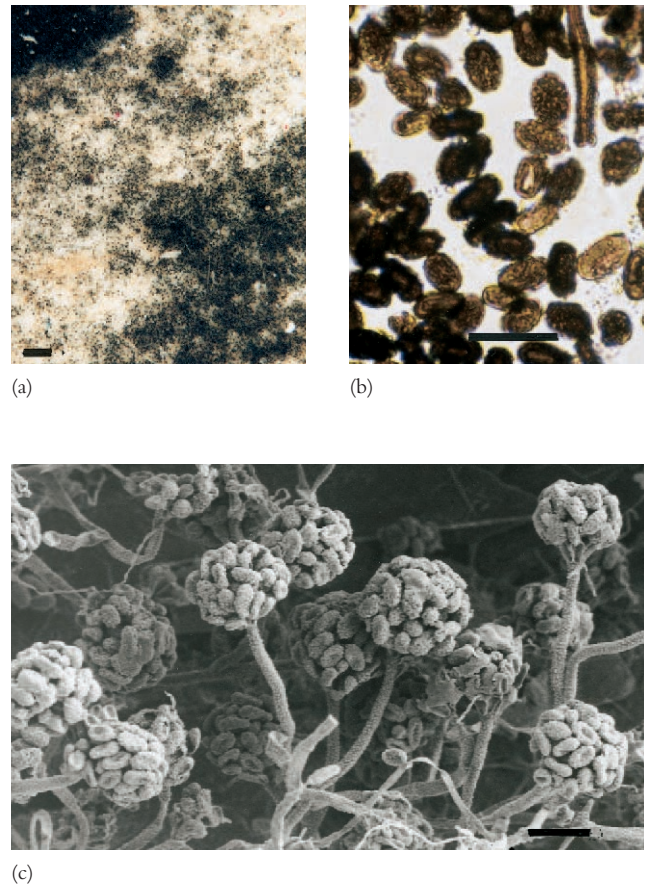


Figure 30.27 Fungal Growth in a Building. Fungal growth on sheet rock removed from a water-damaged building. (a) Stereo microscopic view showing black discolorations. Bar = 500 μ m. (b) Brightfield microscopic observation of conidia similar to *Stachybotrys*, and (c) scanning electron micrograph of dense mycelia and conidiophores characteristic of *Stachybotrys*. Bars = 10 μ m.

30.10 Understanding Microbial Diversity in the Soil

As noted in table 28.8, a wide variety of methods are available to attempt to learn more about the diversity of microorganisms that are present in a soil. As with other environments, such as waters, animal bodies, and insects, it is estimated that only 1 to 10% of the microscopically observable organisms can be cultured in the laboratory. With the availability of molecular techniques, as discussed in chapters 19 and 28, it is now becoming possible to assess the range of microbial diversity in a

soil sample by extracting and analyzing the nucleic acids that are present. Ultimately, it will be necessary to grow the microorganisms and relate specific nucleic acids to known structures in a soil (*see section 28.5*). Without growth of microorganisms considered to contain these nucleic acids, it is not possible to do more than note that these specific sequences are present. This makes it doubly important to better understand the physiology and activities of the microorganisms that make up so much of our world.

1. How can the growth of fungi affect human health inside of a home?
2. What important fungal genera are involved in mold problems in homes?
3. Surface soil spreading of untreated human wastes is carried out in many parts of the world. What are some of the effects of this practice?
4. Why is the growth of microorganisms important for furthering our understanding of microbial diversity?

Summary

1. Terrestrial environments are dominated by the solid phase, consisting of organic and inorganic components.
2. In an ideal soil, microorganisms function in thin water films that have close contact with air (**figure 30.1**). If isolated water volumes are present within the soil volume, from a microbial viewpoint, these are miniaquatic environments.
3. Most microorganisms in these environments are associated with surfaces, and these surfaces influence microbial use of nutrients and interactions with plants and other living organisms (**figure 30.2**).
4. Soils form under a bewildering array of conditions (**figure 30.4**). In all cases organic matter accumulation occurs through the direct activities of primary producers or by the import of preformed organic materials. Soils can be formed in regions such as the antarctic where there are no vascular plants.
5. Bacteria and fungi in soils have different functional strategies. Fungi tend to develop on the surfaces of aggregates, whereas microcolonies of bacteria are commonly associated with smaller pores.
6. Insects and other soil animals are also important parts of the soil. These decomposer-reducers interact with the microorganisms to influence nutrient cycling and other processes.
7. Soils differ in the relationship between organic matter accumulation—primary production—and decomposition.
8. Plants develop associations with many types of microorganisms. These include important associations with rhizobia; actinomycetes, forming actinorhizae; and with endophytic fungi and bacteria. *Agrobacterium* forms tumors and is useful in molecular biology (**figure 30.20**).
9. The *Rhizobium*-legume symbiosis is one of the best-studied examples of plant-microorganism interactions. This interaction is mediated by complex chemicals that serve as communication signals (**figure 30.8**).
10. Mycorrhizal relationships (plant-fungal associations) are varied and complex. Five basic types are now known to exist (**figure 30.11**). The hyphal network of the mycobiont can lead to the formation of a mycorrhizosphere.
11. The mycorrhizal relationship often is established with the assistance of mycorrhization helper bacteria (MHB). In addition, bacteria may occur inside of the mycorrhizal fungus. These bacteria apparently contribute to the nitrogen cycling of the plant-fungus complex.
12. Some plants form tripartite and tetrapartite associations.
13. Composting is useful for maintaining and increasing soil fertility.
14. Soil microorganisms can interact with the atmosphere. Some can serve as nucleating agents, whereas others can degrade airborne pollutants.
15. Microorganisms can play major roles in the dynamics of greenhouse gases such as carbon dioxide, nitrous oxide, nitric oxide, and methane. Microorganisms can contribute to both the production and consumption of these gases.
16. Nitrogen fertilizers can affect soil microbial communities, and the nitrification of ammonium ion-containing fertilizers to nitrate allows the nitrogen to enter ground and surface waters.
17. Fungi play important roles in the decomposition of woody plants. The basidiomycetes are the major decomposers of lignin in woody plant materials. The functioning of these fungi requires the presence of oxygen.
18. Fungi, especially, can produce chemicals that are normally considered as anthropogenic pollutants. These include chloromethane and cyanide.
19. The subsurface includes at least three zones: the shallow subsurface; the zone where gas, oil, and coal have accumulated; and the deep subsurface, where methane synthesis occurs.
20. Microorganisms, particularly the fungi, can develop in moist areas in houses and cause major health problems for humans, including asthma. An important fungus involved in these problems is *Stachybotrys*.

Key Terms

actinorhizae 682
aquifer 691
arbuscular mycorrhizal (AM) fungi 681
arbuscule 681
associative nitrogen fixation 675
bacteroid 676
composting 686
desert crust 673
ectomycorrhiza 681
endomycorrhiza 681
endophyte 679

geosmin 671
greenhouse gas 689
infection thread 676
methanotroph 689
microbivory 672
mycorrhizosphere 681
nitrogen saturation point 686
Nod factors 676
ped 670
phyllosphere 674
residuesphere 690

rhizoplane 675
rhizosphere 675
root nodule 676
slash-and-burn agriculture 672
snow molds 673
stem-nodulating rhizobia 682
subsurface biosphere 691
symbiosome 676
tetrapartite association 685
Ti or Ri plasmid 684
tripartite association 685

Questions for Thought and Review

1. Considering the oxygen gradients, together with nutrient and waste-product gradients present in soils, is there a typical soil microenvironment? Explain.
2. Why might protozoa prefer to use laboratory-grown microorganisms as a nutrient source rather than normal soil microorganisms?
3. Tropical soils throughout the world are under intense pressure in terms of agricultural development. What land use and microbial approaches might be used to better maintain this valuable resource?
4. Microorganisms interact with plants in many ways. How might it be possible to improve these interactions through molecular biology?
5. Why might vascular plants have developed subtle relationships with so many types of microorganisms? What do these molecular-level interactions, which show so many similarities when microbe-plants and microbe-human interactions are considered, suggest concerning possible common evolutionary relationships?
6. Why might a plant assist in its parasitism by *Agrobacterium*?
7. What alternatives could be used to minimize the environmental effects of nitrogen-containing fertilizers on the environment?
8. If lignin-containing plant materials didn't decompose, what could happen?
9. How might you maintain organisms from the deep hot subsurface under their in situ conditions? Compare this problem with that of working with microorganisms from deep marine environments.
10. What are some of the potential strong and weak points of using stable isotope-based analyses of microbial processes?

Critical Thinking Questions

1. How might it be possible to minimize the production of pollutants by soil fungi? Is there any possible role for genetic engineering in solving this problem?
2. Soil bacteria such as *Streptomyces* produce the bulk of known antibiotics. Look up the competitors for *Streptomyces*, the types of antibiotics these bacteria produce, and how the compounds are effective against competitors (what are the physiological targets?). Would you expect aquatic/marine bacteria to be major producers of antibiotics? Why or why not?

Additional Reading

The references provided at the end of chapter 28 also may be consulted for further information.

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PART IX

Nonspecific Resistance and the Immune Response

Chapter 31

Normal Microbiota and Nonspecific
Host Resistance

Chapter 32

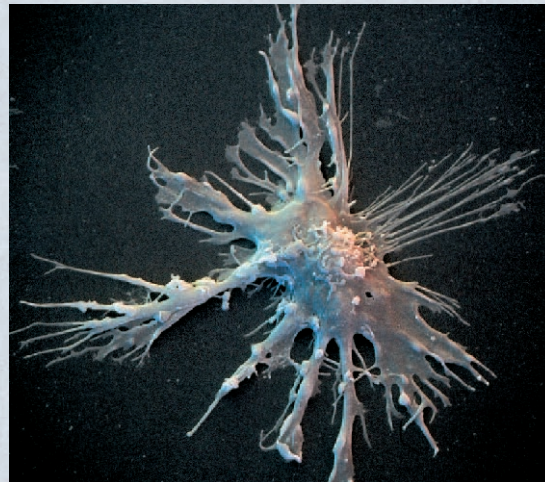
Specific Immunity

Chapter 33

Medical Immunology

CHAPTER 31

Normal Microbiota and Nonspecific Host Resistance



This dendritic cell (electron micrograph) acquired its name because it is covered with long cell extensions that resemble the dendrites of nerve cells. Many dendritic cells process and present antigen to T-helper cells.

Outline

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Concepts

1. Gnotobiotic refers to a microbiologically monitored environment or animal in which the identities of all microorganisms present are known or to an environment or animal that is germfree.
2. Most microorganisms associated with the human body are bacteria; they normally colonize specific sites. There are both positive and negative aspects of these normal microorganisms. Sometimes they compete with pathogens; other times they are capable of producing opportunistic infections.
3. The host's ability to resist infection depends on a constant defense against microbial invasion. Resistance arises from both nonspecific and specific body defense mechanisms.
4. Nonspecific host defenses are those innate mechanisms with which a host is genetically endowed. Examples include physical and mechanical barriers: specifically, the skin, mucous membranes, the respiratory system, gastrointestinal tract, genitourinary tract, and the eye. Other nonspecific host defenses include chemical barriers such as bacteriocins, betalysin, and other polypeptides.
5. Inflammation, the alternative complement pathway, phagocytosis, cytokines, fever, and natural killer cells are other examples of nonspecific defenses that help protect the host against microorganisms and cancer.

Half the secret of resistance is cleanliness, the other half is dirtiness.

—Anonymous

As presented in chapter 28, many microorganisms live much of their lives in a special ecological relationship: an important part of their environment is a member of another species. Previously you learned about many different environments in chapters 28, 29, and 30; this chapter presents the normal microorganisms associated with the human body. Very little is known about the nature of these normal associations, but they are thought to be dynamic interactions rather than associations of mutual indifference. Each member of the symbiosis is thought to derive some benefit from the other, and the associations are for the most part mutualistic or commensalistic (see section 28.2). The survival of a host, such as a human, depends upon an elaborate network of defenses that keeps harmful microorganisms and other foreign material from infecting the body. Should they gain access, additional host defenses are summoned to prevent them from establishing another type of relationship, one of parasitism or **pathogenicity**. Pathogenicity is the ability to produce pathologic changes or disease. A **pathogen** [Greek *patho*, and *gennan*, to produce] is any disease-producing microorganism. Nonspecific host defenses involve general barriers, physical barriers, chemical barriers, and biological barriers such as inflammation and fever. This chapter introduces the normal human microbiota and the main lines of nonspecific host defenses against invasion by harmful microorganisms, cancer cells, and foreign proteins. We begin, however, with a look at the field of gnotobiotics in which it is possible to have an environment or an animal that is free of microorganisms.

31.1 Gnotobiotic Animals

Environments and animals with known microorganisms increasingly are used in research. For example, to determine the role of the normal microorganisms associated with a host, it is possible to deliver an animal by cesarean section and raise that animal in the absence of microorganisms—that is, germfree. These microorganism-free animals provide suitable experimental models for investigating the interactions of animals and their microorganisms. Comparing animals possessing normal microorganisms with germfree animals permits the elucidation of many of the complex relationships covered in chapters 28, 29, and 30 between microorganisms, hosts, and specific environments. Germfree experiments also extend the microbiologist's "pure culture concept" to in vivo research.

The term **gnotobiotic** [Greek *gnotos*, known, and *biota*, the flora and fauna of a region] has been defined in two ways. Some think of a gnotobiotic environment or animal as one in which all the microbiota are known; they distinguish it from one that is truly germfree. We shall use the term in a more inclusive sense. Gnotobiotic refers to a microbiologically monitored environment or animal that is germfree (axenic) or in which the identities of all microbiota are known.

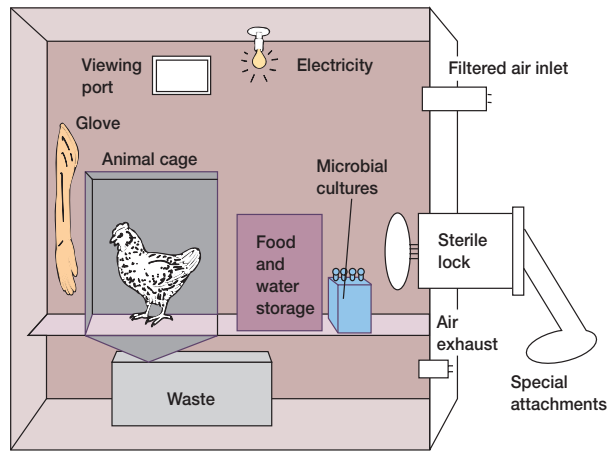
The human fetus in utero (as is the case in most mammals) is usually free from microorganisms. Within hours after birth, it begins to acquire a normal microbiota, which stabilizes during the first week or two of life. From then on, enormous numbers of microorganisms become associated with the human body. Under normal conditions the human lives in symbiosis with billions of microorganisms.

Louis Pasteur first suggested that animals could not live in the absence of microorganisms. Attempts between 1899 and 1908 to grow germfree chickens had limited success because the birds died within a month. Thus it was believed that intestinal bacteria were essential for the adequate nutrition and health of the chickens. It was not until 1912 that germfree chickens were shown to be as healthy as normal birds when they were fed an adequate diet. Since then, gnotobiotic animals and systems have become commonplace in research laboratories (**figure 31.1**).

Gnotobiotic animals and techniques provide good experimental systems to investigate the interactions of animals and specific microorganisms. The comparison of animals containing normal microbiota with gnotobiotic animals helps one better understand the many complex symbiotic associations that exist between the host and its microorganisms.

Establishing a germfree colony of rats, mice, hamsters, rabbits, guinea pigs, or monkeys begins with cesarean sections on pregnant females. The operation is performed under aseptic conditions in a sterile isolator. The newborn animals are then transferred to other germfree isolators in which all entering air, water, and food is sterile. Once the gnotobiotic animals become established, normal mating among themselves maintains the germfree colony.

Establishment of a germfree colony is much easier with chickens and other birds than with mammals. Fertile eggs from the birds are first sterilized with a germicide and then placed in



(a)



(b)

Figure 31.1 Raising Gnotobiotic Animals. (a) Schematic of a gnotobiotic isolator. The microbiological cultures monitor the sterile environment. If growth occurs on any of the cultures, gnotobiotic conditions do not exist. (b) Gnotobiotic isolators for rearing colonies of small mammals.

sterile isolators. When the chick hatches, it is sterile and able to feed by itself. To make sure that a colony is germfree, periodic bacteriologic examination of all exhaust air, waste material, and cages must be done. Microorganisms should not be found.

Germfree animals are not anatomically and physiologically normal. For example, they possess poorly developed lymphoid tissue, a thin intestinal wall, an enlarged cecum (where one is present), and a low antibody titer (see section 32.3). They require high amounts of vitamin K and the B complexes. Germfree animals also have reduced cardiac output and lower metabolic rates.

Germfree animals are usually more susceptible to pathogens. With the normal protective microbiota absent, foreign and pathogenic microorganisms establish themselves very easily. The number of microorganisms necessary to infect a germfree animal and produce a diseased state also is much smaller. Conversely, germfree animals are almost completely resistant to the intestinal protozoan *Entamoeba histolytica* that causes amebic dysentery. This resistance results from the absence of the bacteria that *E. histolytica* uses as a food source. Germfree animals also do not show any dental caries or plaque formation (see section 39.6). However, if they are inoculated with cariogenic (caries or cavity-causing) streptococci of the *Streptococcus mutans*–*Streptococcus gordonii* group and fed a high-sucrose diet, they will develop caries. (*S. gordonii* was formerly considered a subpopulation of *S. sanguis*.)

Entamoeba histolytica (pp. 950–51)

1. Define gnotobiotic.
2. How would you establish a germfree colony of mice? Of chickens?
3. Compare a germfree mouse to a normal one with regard to overall health. What benefits does an animal gain from its microbiota?

31.2 Normal Microbiota of the Human Body

In a healthy human the internal tissues (e.g., brain, blood, cerebrospinal fluid, muscles) are normally free of microorganisms. Conversely, the surface tissues (e.g., skin and mucous membranes) are constantly in contact with environmental microorganisms and become readily colonized by certain microbial species. The mixture of microorganisms regularly found at any anatomical site is referred to as the normal microbiota, the indigenous microbial population, the microflora, or the normal flora. For consistency, the term **normal microbiota** is used in this chapter. An overview of the microbiota native to different regions of the body (figure 31.2) and an introduction to the microorganisms one can expect to find on culture reports is presented next. Because bacteria make up most of the normal microbiota, they are emphasized over the fungi (mainly yeasts) and protozoa.

There are many reasons to acquire knowledge of the normal human microbiota. Four specific examples include:

1. An understanding of the different microorganisms at specific locations provides greater insight into the possible infections that might result from injury to these body sites.
2. A knowledge of the normal microbiota in an infected part of the body gives the physician-investigator a better perspective concerning the possible source and significance of microorganisms isolated from an infection site.
3. A knowledge of the normal microbiota helps the physician-investigator understand the causes and consequences of colonization and growth by microorganisms normally absent at a specific body site.
4. An increased awareness of the role that these normal microbiota play in stimulating the host immune response can be gained. This awareness is important because the immune system provides protection against potential pathogens.

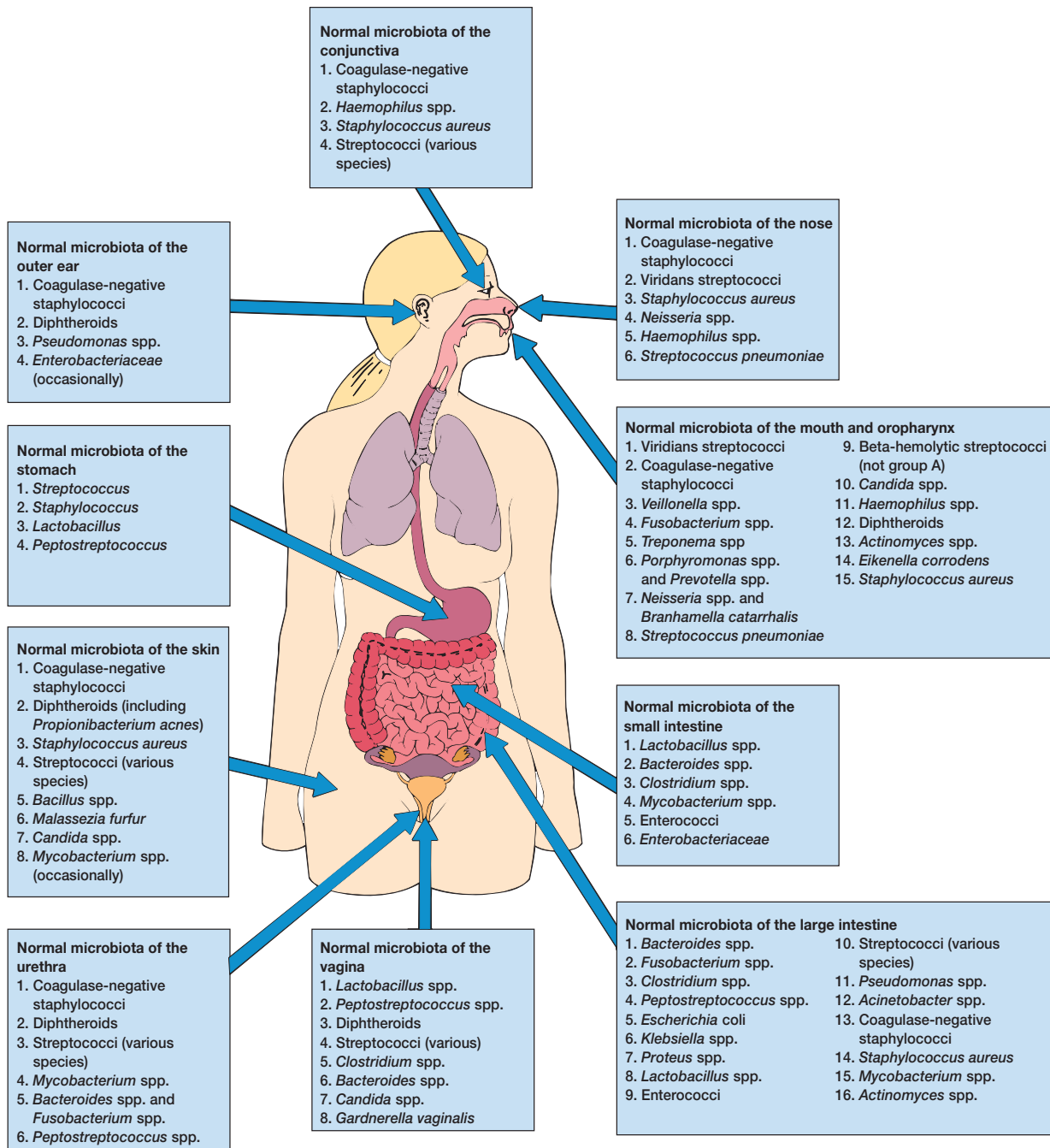


Figure 31.2 Normal Microbiota of a Human. A compilation of microorganisms that constitute normal microbiota encountered in various body sites.

Distribution of the Normal Microbiota

As noted in chapter 28, three of the most important types of symbiotic relationships are commensalism, mutualism, and parasitism. Within each category the association may be either ectosymbiotic or endosymbiotic. In **ectosymbiosis** one organism remains outside the other. In **endosymbiosis** one organism is present within the other. In the following subsections, examples will be presented of both ecto- and endosymbiotic relationships that exist with the human host. Both commensalistic and mutualistic relationships are considered. Parasitism and pathogenicity will be presented in chapter 34.

Skin

The adult human is covered with approximately 2 square meters of skin. It has been estimated that this surface area supports about 10^{12} bacteria. As noted in chapter 28, a commensal is an organism that participates in commensalism. Commensalism is a symbiotic relationship in which one species benefits and the other is unharmed. Commensal microorganisms living on or in the skin can be either resident (normal) or transient microbiota. Resident organisms normally grow on or in the skin. Their presence becomes fixed in well-defined distribution patterns. Those microorganisms that are temporarily present are transients. Transients usually do not become firmly entrenched; they are unable to multiply and normally die in a short time.

The anatomy and physiology of the skin vary from one part of the body to another, and the normal resident microbiota reflect these variations. The skin surface or epidermis is not a favorable environment for microbial colonization. Several factors are responsible for this hostile microenvironment. First, the skin is subject to periodic drying. Lack of moisture drives many resident microbiota into a dormant state. However, in certain parts of the body (scalp, ears, axillary areas, genitourinary and anal regions, perineum, palms), moisture is sufficiently high to support a resident microbiota. Second, the skin has a slightly acidic pH due to the organic acids produced by normal staphylococci and secretions from skin oil and sweat glands. The acidic pH (4 to 6) discourages colonization by many microorganisms. Third, sweat contains a high concentration of sodium chloride. This makes the skin surface hyperosmotic and osmotically stresses most microorganisms. Finally, certain inhibitory substances (bactericidal and/or bacteriostatic) on the skin help control colonization, overgrowth, and infection from microorganisms. For example, the sweat glands release lysozyme (muramidase), an enzyme that lyses *Staphylococcus epidermidis* and other gram-positive bacteria by hydrolyzing the β (1 $\rightarrow$ 4) glycosidic bond connecting *N*-acetylmuramic acid and *N*-acetylglucosamine in the bacterial cell wall peptidoglycan (figure 31.9). The oil glands secrete complex lipids that may be partially degraded by the enzymes from certain gram-positive bacteria (*Propionibacterium acnes*). These bacteria can change the secreted lipids to unsaturated fatty acids such as oleic acid that have strong antimicrobial activity against gram-negative bacteria and some fungi. Some of these fatty acids are volatile and may be associated with a strong odor. Therefore many body deodorants

contain antibacterial substances that act selectively against gram-positive bacteria to reduce the production of volatile unsaturated fatty acids and body odor. However, deodorants can shift the normal microbiota to predominantly gram-negative bacteria and precipitate subsequent infections.

Most skin bacteria are found on the superficial cells, colonizing dead cells, or closely associated with the oil and sweat glands. Secretions from these glands provide the water, amino acids, urea, electrolytes, and specific fatty acids that serve as nutrients primarily for *Staphylococcus epidermidis* and aerobic corynebacteria. Gram-negative bacteria generally are found in the moister regions. The yeasts *Pityrosporum ovale* and *P. orbiculare* normally occur on the scalp. Some dermatophytic fungi may colonize the skin and produce athlete's foot and ringworm. [Athlete's foot disease \(p. 944\)](#)

The most prevalent bacterium in the skin glands is the gram-positive, anaerobic, lipophilic rod *Propionibacterium acnes*. This bacterium usually is harmless; however, it has been associated with the skin disease acne vulgaris. Acne commonly occurs during adolescence when the endocrine system is very active. Hormonal activity stimulates an overproduction of **sebum**, a fluid secreted by the oil glands. A large volume of sebum accumulates within the glands and provides an ideal microenvironment for *P. acnes*. In some individuals this accumulation triggers an inflammatory response that causes redness and swelling of the gland's duct and produces a **comedo** [pl., comedones], a plug of sebum and keratin in the duct. Inflammatory lesions (papules, pustules, nodules) commonly called "blackheads" or "pimples" can result. *P. acnes* produces lipases that hydrolyse the sebum triglycerides into free fatty acids. Free fatty acids are especially irritating because they can enter the dermis and promote inflammation. Because *P. acnes* is extremely sensitive to tetracycline, this antibiotic may aid acne sufferers. Accutane, a synthetic form of vitamin A, is also used.

Some pathogens found on or in the skin are residents that colonize the area around orifices. *Staphylococcus aureus* is the best example. It resides in the nostrils and perianal region but survives poorly elsewhere. In like manner *Clostridium perfringens* usually colonizes only the perineum and thighs, especially in those who suffer from diabetes.

1. What are four reasons why knowledge of the normal human microbiota is important?
2. Why is the skin not usually a favorable microenvironment for colonization by bacteria?
3. How do microorganisms contribute to body odor?
4. What physiological role does *Propionibacterium acnes* play in the establishment of acne vulgaris?

Nose and Nasopharynx

The normal microbiota of the nose is found just inside the nostrils. *Staphylococcus aureus* and *S. epidermidis* are the predominant bacteria present and are found in approximately the same numbers as on the skin of the face.

The nasopharynx, that part of the pharynx lying above the level of the soft palate, may contain small numbers of potentially pathogenic bacteria such as *Streptococcus pneumoniae*, *Neisseria meningitidis*, and *Haemophilus influenzae*. Diphtheroids, a large group of nonpathogenic gram-positive bacteria that resemble *Corynebacterium* (see section 24.5), are commonly found in both the nose and nasopharynx.

Oropharynx

The oropharynx is that division of the pharynx lying between the soft palate and the upper edge of the epiglottis. Like the nose, large numbers of *Staphylococcus aureus* and *S. epidermidis* inhabit this region. The most important bacteria found in the oropharynx are the various alpha-hemolytic streptococci (*S. oralis*, *S. milleri*, *S. gordonii*, *S. salivarius*); large numbers of diphtheroids; *Branhamella catarrhalis*; and small gram-negative cocci related to *Neisseria meningitidis*. It should be noted that the palatine and pharyngeal tonsils harbor a similar microbiota, except that within the tonsillar crypts, there is an increase in *Micrococcus* and the anaerobes *Porphyromonas*, *Prevotella*, and *Fusobacterium*. (*Porphyromonas* spp. and *Prevotella* spp. were formerly classified as *Bacteroides*.)

Respiratory Tract

The upper and lower respiratory tracts (trachea, bronchi, bronchioles, alveoli) do not have a normal microbiota. This is because microorganisms are removed by (1) the continuous stream of mucus generated by the ciliated epithelial cells and (2) the phagocytic action of the alveolar macrophages. In addition, a bactericidal effect is exerted by the enzyme lysozyme, present in nasal mucus. [Mucociliary blanket \(p. 711\)](#)

Mouth

The normal microbiota of the mouth or oral cavity contains organisms able to resist mechanical removal by adhering to surfaces like the gums and teeth. Those that cannot attach are removed by the mechanical flushing of the oral cavity contents to the stomach where they are destroyed by hydrochloric acid. The continuous desquamation (shedding) of epithelial cells also removes microorganisms. Those microorganisms able to colonize the mouth find a very comfortable environment due to the availability of water and nutrients, the suitability of pH and temperature, and the presence of many other growth factors. [Biofilms \(pp. 620–22\)](#)

The oral cavity is colonized by microorganisms from the surrounding environment within hours after a human is born. Initially the microbiota consists mostly of the genera *Streptococcus*, *Neisseria*, *Actinomyces*, *Veillonella*, and *Lactobacillus*. Some yeasts also are present. Most microorganisms that invade the oral cavity initially are aerobes and obligate anaerobes. When the first teeth erupt, the anaerobes (*Porphyromonas*, *Prevotella*, and *Fusobacterium*) become dominant due to the anaerobic nature of the space between the teeth and gums. As the teeth grow, *Streptococcus parasanguis* and *S. mutans* attach to their enamel surfaces; *S. sali-*

varius attaches to the buccal and gingival epithelial surfaces and colonizes the saliva. These streptococci produce a glycocalyx and various other adherence factors that enable them to attach to oral surfaces. The presence of these bacteria contributes to the eventual formation of dental plaque, caries, gingivitis, and periodontal disease. [Periodontal disease \(p. 936\)](#)

Eye

At birth and throughout human life, a small number of bacterial commensals are found on the conjunctiva of the eye. The predominant bacterium is *Staphylococcus epidermidis* followed by *S. aureus*, aerobic corynebacteria (diphtheroids), and *Streptococcus pneumoniae*. Cultures from the eyelids or conjunctiva also yield *Branhamella catarrhalis*, *Escherichia*, *Klebsiella*, *Proteus*, *Enterobacter*, *Neisseria*, and *Bacillus* species. Few anaerobic organisms are present.

External Ear

The normal microbiota of the external ear resemble those of the skin, with coagulase-negative staphylococci and *Corynebacterium* predominating. Less frequently found are *Bacillus*, *Micrococcus*, and *Neisseria* species. Gram-negative rods such as *Proteus*, *Escherichia*, and *Pseudomonas* are occasionally seen. Mycological studies show the following fungi to be normal microbiota: *Aspergillus*, *Alternaria*, *Penicillium*, *Candida*, and *Saccharomyces*.

Stomach

As noted earlier, many microorganisms are washed from the mouth into the stomach. Owing to the very acidic pH values (2 to 3) of the gastric contents, most microorganisms are killed. As a result the stomach usually contains less than 10 viable bacteria per milliliter of gastric fluid. These are mainly *Sarcina*, *Streptococcus*, *Staphylococcus*, *Lactobacillus*, *Peptostreptococcus*, and yeasts such as *Candida* spp. Microorganisms may survive if they pass rapidly through the stomach or if the organisms ingested with food are particularly resistant to gastric pH (mycobacteria). Normally the number of microorganisms increases after a meal but quickly falls as the acidic pH takes its toll. Changes in the gastric microbiota also occur if there is an increase in gastric pH following intestinal obstruction, which permits a reflux of alkaline duodenal secretions into the stomach. If the gastric pH increases, the microbiota of the stomach are likely to reflect that of the oropharynx and, in addition, contain both gram-negative aerobic and anaerobic bacteria.

Small Intestine

The small intestine is divided into three anatomical areas: the duodenum, jejunum, and ileum. The duodenum (the first 25 cm of the small intestine) contains few microorganisms because of the combined influence of the stomach's acidic juices and the inhibitory action of bile and pancreatic secretions. Of the bacteria present, gram-positive cocci and rods comprise most of the microbiota. *Enterococcus faecalis*, lactobacilli, diphtheroids, and the yeast *Candida albicans* are occasionally found in the je-

Box 31.1

Probiotics for Humans and Animals

The large intestine of humans and animals contains a very complex and balanced microbiota. These microorganisms normally prevent infection and have a positive effect on nutrition. Any abrupt change in diet, stress, or antibiotic therapy can upset this microbial balance, making the host susceptible to disease and decreasing the efficiency of food use. **Probiotics** [Greek *pro*, for, and *bios*, life], the oral administration of either living microorganisms or substances to promote health and growth, has the potential to reestablish the natural balance and return the host to normal health and nutrition.

Probiotic microorganisms are host specific; thus a strain selected as a probiotic in one animal may not be suitable in another species. Furthermore, microorganisms selected for probiotic use should exhibit the following characteristics:

1. Adhere to the intestinal mucosa of the host
2. Be easily cultured
3. Be nontoxic and nonpathogenic to the host
4. Exert a beneficial effect on the host
5. Produce useful enzymes or physiological end products that the host can use
6. Remain viable for a long time
7. Withstand HCl in the host's stomach and bile salts in the small intestine

There are several possible explanations of how probiotic microorganisms displace pathogens and enhance the development and stability of the microbial balance in the large intestine. These include:

1. Competition with pathogens for nutrients and adhesion sites
2. Inactivation of pathogenic bacterial toxins or metabolites
3. Production of substances that inhibit pathogen growth
4. Stimulation of nonspecific immunity

A wide variety of probiotic preparations have been patented for cattle, goats, horses, pigs, poultry, sheep, and domestic animals. Most of these preparations contain lactobacilli and/or streptococci; a few contain bifidobacteria.

Evidence has accumulated that certain probiotic microorganisms also offer considerable health benefits for humans. Potential benefits include:

1. Anticarcinogenic activity
2. Control of intestinal pathogens
3. Improvement of lactose use in individuals who have lactose intolerance
4. Reduction in the serum cholesterol concentration

Although the development of probiotics is still in its early stages, a better understanding of the normal microbiota in the large intestine of both animals and humans will be forthcoming as more microbiologists investigate probiotic activity. [Food microbiology \(chapter 41\)](#)

junum. In the distal portion of the small intestine (ileum), the microbiota begin to take on the characteristics of the colon microbiota. It is within the ileum that the pH becomes more alkaline. As a result anaerobic gram-negative bacteria and members of the family *Enterobacteriaceae* become established.

Large Intestine (Colon)

The large intestine or colon has the largest microbial community in the body. Microscopic counts of feces approach 10^{12} organisms per gram wet weight. Over 400 different species have been isolated from human feces. The colon can be viewed as a large fermentation vessel, and the microbiota consist primarily of anaerobic, gram-negative, nonsporing bacteria and gram-positive, spore-forming, and nonsporing rods. Not only are the vast majority of microorganisms anaerobic, but many different species are present in large numbers. Several studies have shown that the ratio of anaerobic to facultative anaerobic bacteria is approximately 300 to 1. Even the most abundant of the latter, *Escherichia coli*, is only about 0.1% of the total population.

Besides the many bacteria in the large intestine, the yeast *Candida albicans* and certain protozoa may occur as harmless commensals. *Trichomonas hominis*, *Entamoeba hartmanni*, *Endolimax nana*, and *Iodamoeba butschlii* are common inhabitants.

[Protozoan diseases \(pp. 950–58\)](#)

Various physiological processes move the microbiota through the colon so an adult eliminates about 3×10^{13} microorganisms daily. These processes include peristalsis and desquamation of the surface epithelial cells to which microorganisms are attached, and continuous flow of mucus that carries adhering microorganisms with it. To maintain homeostasis of the microbiota, the body must continually replace those lost microorganisms. The bacterial population in the human colon usually doubles once or twice a day. Under normal conditions the resident microbial community is self-regulating. Competition and mutualism between different microorganisms and between the microorganisms and their host serve to maintain a status quo. However, if the intestinal environment is disturbed, the normal microbiota may change greatly. Disruptive factors include stress, altitude changes, starvation, parasitic organisms, diarrhea, and use of antibiotics or probiotics (**Box 31.1**). Finally, it should be emphasized that the actual proportions of the individual bacterial populations within the indigenous microbiota depend largely on a person's diet.

The initial residents of the colon of breast-fed infants are members of the gram-positive *Bifidobacterium* genus, because human milk contains a disaccharide amino sugar that *Bifidobacterium* species require as a growth factor. In formula-fed infants, gram-positive *Lactobacillus* species predominate because formula lacks the required growth factor. With the ingestion of solid food, these initial colonizers of the colon are eventually displaced

by a typical gram-negative microbiota. Ultimately the composition of the adult's microbiota is established.

Genitourinary Tract

The upper genitourinary tract (kidneys, ureters, and urinary bladder) is usually free of microorganisms. In both the male and female, a few bacteria (*Staphylococcus epidermidis*, *Enterococcus faecalis*, and *Corynebacterium* spp.) usually are present in the distal portion of the urethra. *Neisseria* and some members of the *Enterobacteriaceae* are occasionally found.

In contrast, the adult female genital tract, because of its large surface area and mucous secretions, has a complex microbiota that constantly changes with the female's menstrual cycle. The major microorganisms are the acid-tolerant lactobacilli, primarily *Lactobacillus acidophilus*, often called Döderlein's bacilli. They ferment the glycogen produced by the vaginal epithelium, forming lactic acid. As a result the pH of the vagina and cervix is maintained between 4.4 and 4.6.

1. What are the most common microorganisms found in the nose? The oropharynx? The nasopharynx? The tonsillar crypts? The lower respiratory tract? The mouth? The eye? The external ear? The stomach? The small intestine? The colon? The genitourinary tract?
2. Why is the colon considered a large fermentation vessel?
3. What physiological processes move the microbiota through the gastrointestinal tract?
4. How do the initial microbial colonizers of breast-fed infants differ from those of bottle-fed infants?
5. Describe the microbiota of the upper and lower female genitourinary tract.

The Relationship between Normal Microbiota and the Host

The interaction between a host and a microorganism is a dynamic process in which each protagonist acts to maximize its survival. In some instances, after a microorganism enters or contacts a host, a positive mutually beneficial relationship occurs that becomes integral to the health of the host. These microorganisms become the normal microbiota. In other instances, the microorganism produces or induces deleterious effects on in the host; the end result may be disease or even death of the host (see chapter 34).

Our environment is teeming with microorganisms and we come in contact with many of them every day. Some of these microorganisms are pathogenic—that is, they cause disease. Yet these pathogens are at times prevented from causing disease by the competition provided by the normal microbiota. In general, the normal microbiota use space, resources, nutrients, and may produce chemicals that repel invading pathogens. These normal microbiota prevent colonization of pathogens and possible disease through “bacterial interference.” For instance, the lactobacilli in the female genital tract maintain a low pH and inhibit colonization by pathogenic bacteria, and the corynebacteria on the skin produce fatty acids that inhibit col-

onization by pathogenic bacteria. This is an excellent example of amensalism (see p. 609).

Although normal microbiota offer some protection from invading pathogens, they may themselves become pathogenic and produce disease under certain circumstances, and then are termed **opportunistic microorganisms** or **pathogens**. These opportunistic microorganisms are adapted to the noninvasive mode of life defined by the limitations of the environment in which they are living. If removed from these environmental restrictions and introduced into the bloodstream or tissues, disease can result. For example, streptococci of the viridans group are the most common resident bacteria of the upper respiratory tract. If large numbers of them are introduced into the bloodstream (e.g., following tooth extraction or a tonsillectomy), they may settle on deformed or prosthetic heart valves and cause endocarditis.

Opportunistic microorganisms often cause disease in compromised hosts. A **compromised host** is seriously debilitated and has a lowered resistance to infection. There are many causes of this condition including malnutrition, alcoholism, cancer, diabetes, leukemia, another infectious disease, trauma from surgery or an injury, an altered normal microbiota from the prolonged use of antibiotics, and immunosuppression by various factors (e.g., drugs, viruses [HIV], hormones, and genetic deficiencies). For example, *Bacteroides* species are the most common residents in the large intestine (figure 31.2) and are quite harmless in that location. If introduced into the peritoneal cavity or into the pelvic tissue as a result of trauma, they cause suppuration (the formation of pus) and bacteremia (the presence of bacteria in the blood). Many other examples of opportunistic infections will be presented in chapters 38, 39, and 40. The important point here is that the normal microbiota are harmless and may be beneficial in their normal location in the host and in the absence of coincident abnormalities. However, they can produce disease if introduced into foreign locations or compromised hosts.

1. Give one example of the normal microbiota benefitting a host.
2. Provide two examples of how the normal host microbiota prevent the establishment of a pathogen.
3. How would you define an opportunistic microorganism or pathogen? A compromised host?

31.3 Overview of Host Resistance

To establish an infection, the pathogenic microorganism must first overcome many surface barriers, such as enzymes and mucus, that are either directly antimicrobial or inhibit attachment of the microorganism to the host (section 31.5). Because neither the surface of the skin nor the mucus-lined body cavities are ideal environments for many microorganisms, some pathogens must breach these barriers and get to underlying tissues. Any microorganism that penetrates these barriers encounters the two levels of resistance: other nonspecific resistance mechanisms and the specific immune response.

Vertebrates (including humans) are continuously exposed to microorganisms and their metabolic products that can cause disease. Fortunately these animals are equipped with an immune sys-

tem that protects them against adverse consequences of this exposure. The **immune system** is composed of widely distributed cells, tissues, and organs that recognize foreign substances and microorganisms and act to neutralize or destroy them. **Immunity** [Latin *immunis*, free of burden] refers to the general ability of a host to resist a particular infection or disease. **Immunology** is the science that is concerned with immune responses to the foreign challenge and how these responses are used to resist infection. It includes the distinction between “self” and “nonself” and all the biological, chemical, and physical aspects of the immune response.

There are two fundamentally different types of immune responses to invading microorganisms and foreign material. The **nonspecific immune response** is also known as **nonspecific resistance** and **innate** or **natural immunity**; it offers resistance to any microorganism or foreign material encountered by the vertebrate host. It includes general mechanisms inherited as part of the innate structure and function of each animal, and acts as a first line of defense. The nonspecific immune response lacks immunological memory—that is, nonspecific responses occur to the same extent each time a microorganism or foreign body is encountered.

In contrast, the **specific immune responses** also known as **acquired** or **specific immunity** resist a particular foreign agent; moreover, specific immune responses improve on repeated exposure to foreign agents such as viruses, bacteria, and toxins. Substances that are recognized as foreign and provoke immune responses are called antigens. The antigens cause specific cells to produce proteins called antibodies. Antibodies bind to and inactivate a specific antigen. Other cells destroy virus-infected cells. The nonspecific and specific responses usually work together to eliminate pathogenic microorganisms and other foreign agents.

Section 31.4 provides an overview of the participants in the immune system, and section 31.5 describes the nonspecific immune responses. The specific immune response will be covered in chapter 32.

1. Define each of the following terms: immune system, immunity, immunology, antigen, antibody.
2. Briefly describe the nonspecific immune response. What are some of its characteristics?
3. Briefly describe the specific immune response. What are some of its characteristics?

31.4 Cells, Tissues, and Organs of the Immune System

The immune system is an organization of cells, tissues, organs, and molecules with specialized roles in defending against viruses, microorganisms, cancer cells, and nonself proteins (e.g., organ transplants). Immune system cells and tissue will now be considered.

Cells of the Immune System

The cells responsible for both nonspecific and specific immunity are the white blood cells called **leukocytes**. [Greek *leukos*, white,

and *kytos*, cell]. All of the leukocytes originate from pluripotent stem cells in the fetal liver and in the bone marrow of the animal host (**figure 31.3**), from which they migrate to other body sites, undergo further development, and perform their various functions. These cells of the immune system are present throughout the host's body. Some become residents within tissues, where they respond to local trauma and sound the alarm; others circulate in body fluids and are recruited to the sites of infection. In defending the host against pathogenic microorganisms, leukocytes cooperate with each other first to recognize the pathogen as an invader and then to destroy it. These different leukocytes are now briefly examined.

Lymphoid Cells

Lymphocytes [Latin *lympa*, water, and *cyte*, cell] are the major cells of the specific immune system. Lymphocytes can be divided into three populations: T cells, B cells, and natural killer cells. **B cells** or **B lymphocytes** reach maturity within the bone marrow, circulate in the blood, and also settle in various lymphoid organs. **T cells** or **T lymphocytes** mature in the thymus gland; they can remain in the thymus, circulate in the blood, or reside in lymphoid organs such as the lymph nodes and spleen. B cells and T cells will be discussed further in chapter 32 along with their roles in specific immunity. Natural killer cells are important in killing cells infected with either viruses or intracellular bacteria and destroying cancer cells (section 31.10).

Mononuclear Cells

There are two types of mononuclear (i.e., a single large nucleus) cells—monocytes and macrophages. Both types are highly phagocytic and make up the **monocyte-macrophage system** (**figure 31.4**). Recall (*see figure 4.10*) that during phagocytosis large particles and even other microorganisms are engulfed and enclosed in a phagocytic vacuole or phagosome.

Monocytes [Greek *monos*, single, and *cyte*, cell] are mononuclear phagocytic leukocytes with an ovoid or kidney-shaped nucleus and granules in the cytoplasm that stain gray-blue (**figure 31.3**). They are produced in the bone marrow and enter the blood, circulate for about eight hours, enlarge, migrate to the tissues, and mature into macrophages.

As just noted, **macrophages** [Greek *macros*, large, and *phagein*, to eat] are derived from monocytes and are also classified as mononuclear phagocytic leukocytes. However, they may be larger than monocytes, contain more organelles (especially lysosomes and phagolysosomes), and have a plasma membrane covered with ruffles or microvilli (**figure 31.5**). Macrophages have receptors for antibodies and complement; these can coat microorganisms or foreign material and enhance phagocytosis. This enhancement is termed opsonization. Macrophages spread throughout the animal body and take up residence in specific tissues where they are given special names (**figure 31.4**). Since macrophages are highly phagocytic, their function in nonspecific resistance will be discussed in more detail in the context of phagocytosis.

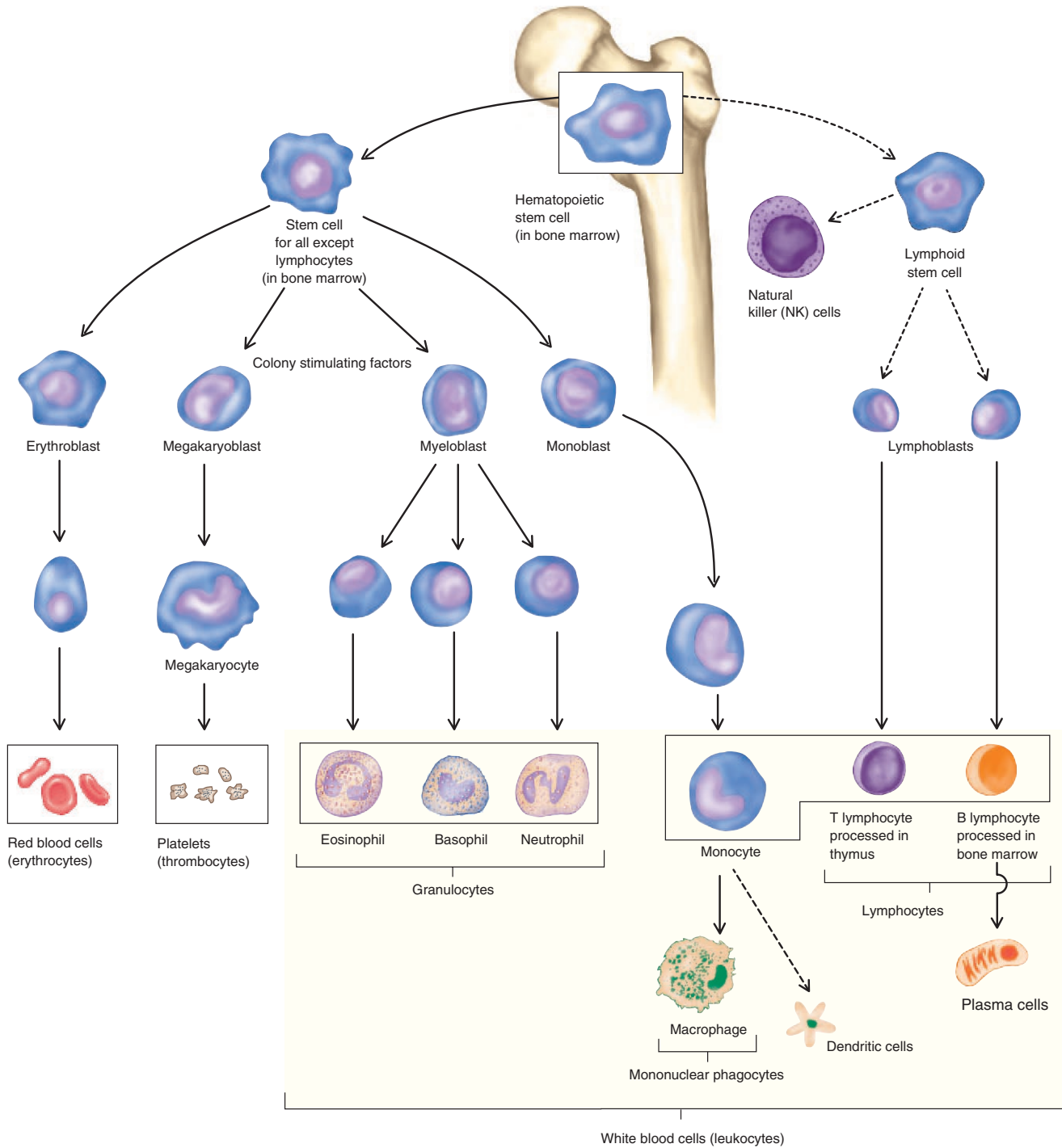


Figure 31.3 The Different Types of Human Blood Cells. Pluripotent stem cells in the bone marrow divide to form two lineages: (1) the lymphoid stem cell gives rise to B cells that become antibody-secreting plasma cells, T cells that become activated T cells, and natural killer cells. (2) The common myeloid progenitor cell gives rise to the granulocytes (neutrophils, eosinophils, basophils), monocytes that give rise to macrophages and dendritic cells, an unknown precursor that gives rise to mast cells, megakaryocytes that produce platelets, and the erythroblast that produces erythrocytes (red blood cells). The dashed lines indicate steps that have not been very well studied.

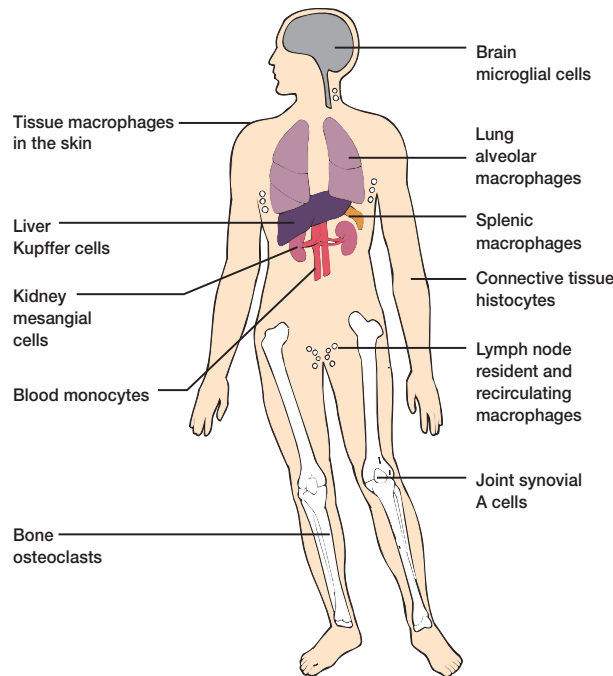


Figure 31.4 The Monocyte-Macrophage System. This system consists of tissue (such as found within the liver, spleen, and lymph nodes) containing “fixed” or immobile phagocytes that have specific names depending on their location.

Granulocytes

Granulocytes have irregular-shaped nuclei with two to five lobes, and the cytoplasmic matrix has granules (figure 31.3) that contain reactive substances that kill microorganisms and enhance inflammation. Because of the irregular-shaped nuclei, granulocytes are also called **polymorphonuclear leukocytes** or **PMNs**. Three types of granulocytes exist: basophils, eosinophils, and neutrophils.

Basophils [Greek *basis*, base, and *philein*, to love] have an irregular-shaped nucleus with two lobes, and the granules stain bluish-black with basic dyes. Basophils are nonphagocytic cells that function by releasing histamine, prostaglandins, serotonin, and leukotrienes from their granules upon appropriate stimulation. Because these physiological mediators influence the tone and diameter of blood vessels, they are termed vasoactive agents. Basophils (and mast cells) possess high-affinity receptors for an immunoglobulin (IgE; see figure 32.14) and thereby become coated with IgE antibodies. Once coated, antigens can trigger the cell to secrete vasoactive mediators. These inflammatory mediators play a major role in certain allergic responses such as eczema, hay fever, and asthma.

Eosinophils [Greek *eos*, dawn, and *philiēn*] have a two-lobed nucleus connected by a slender thread of chromatin, and the granules stain red with acidic dyes. Unlike basophils, eosinophils



Figure 31.5. Phagocytosis by a Macrophage. One type of nonspecific host resistance involves white blood cells called macrophages and the process of phagocytosis. This scanning electron micrograph ($\times 3,000$) shows a macrophage devouring a colony of bacteria. Phagocytosis is but one of many nonspecific defenses humans and other animals have to combat microbial pathogens.

are mobile cells that can migrate from the bloodstream into tissue spaces. Their role is important only in the defense against protozoan and helminth parasites, mainly by releasing cationic proteins and reactive oxygen metabolites into the extracellular fluid to damage the parasite’s plasma membranes.

Neutrophils [Latin *neuter*, neither, and *philiēn*] stain readily at a neutral pH, have a nucleus with three to five lobes connected by slender threads of chromatin, and contain fine primary and secondary inconspicuous granules. Like macrophages, neutrophils have receptors for antibodies and complement proteins and are highly phagocytic cells. However, unlike the macrophage, neutrophils do not reside in healthy tissue but rapidly migrate to the site of tissue damage and infection where they are the principal phagocytic and microbicidal cells. The lytic enzymes and bactericidal substances in neutrophils are contained within large primary and smaller secondary granules. Primary granules contain peroxidase, lysozyme, and various hydrolytic enzymes, whereas secondary granules have collagenase, lactoferrin, and lysozyme. Both of these granules help accomplish intracellular digestion. Neutrophils also use oxygen-dependent and oxygen-independent pathways that generate antimicrobial substances and defensins to kill ingested microorganisms (section 31.8). Neutrophils are described in more detail in the contexts of the inflammatory response and phagocytosis.

Mast Cells

Mast cells are bone marrow-derived cells found in connective tissue. They contain granules with histamine and other pharmacologically active substances that contribute to the inflammatory

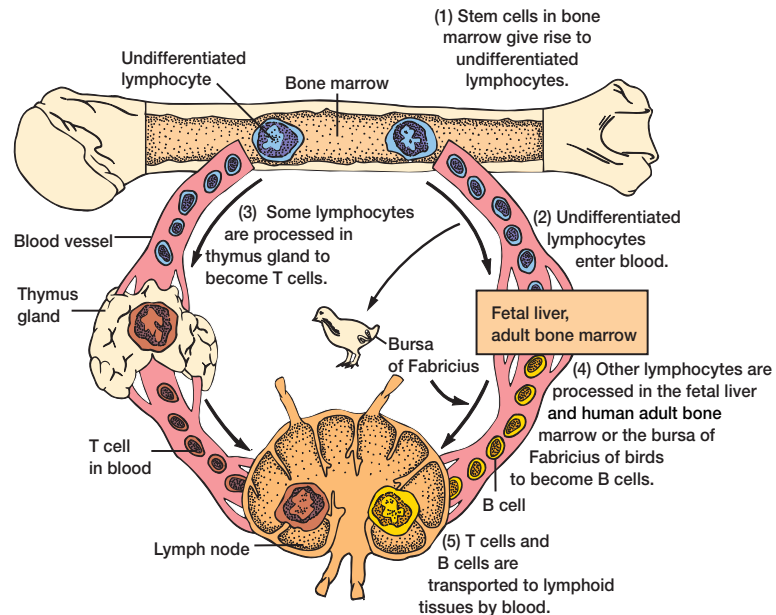


Figure 31.6 Schematic Representation of Lymphocyte Development. Bone marrow releases undifferentiated lymphocytes, which, after processing, become T and B cells.

response. Mast cells, along with basophils, play an important role in the development of allergies and hypersensitivities (*see section 33.2*).

Dendritic Cells

Dendritic cells (chapter opener figure, p. 697) can recognize specific pathogen-associated molecular patterns on microorganisms and play an important role in nonspecific resistance. They can differentiate between potentially harmful microorganisms and “self” molecules. After the pathogen is recognized, it binds to the dendritic cell’s pattern recognition receptors and then is phagocytosed. These cells are also stimulated by endogenous activators such as interferon- α , heat-shock proteins, and tumor necrosis factor that are released in response to microbial infection. After stimulation, dendritic cells migrate to the bloodstream or lymphatic system and present antigens to T cells. Thus dendritic cells also play an important role in the specific immune response.

Organs and Tissues of the Immune System

Based on function, the organs and tissues of the immune system can be divided into primary or secondary lymphoid organs or tissues. The primary organs or tissues are where immature lymphocytes mature and differentiate into antigen-sensitive mature B and T cells. The thymus is the primary lymphoid organ and bone marrow is the primary lymphoid tissue. The secondary organs and tissues serve as areas where lymphocytes may encounter and bind antigen, whereupon they proliferate and differentiate into fully mature, antigen-specific effector cells. The spleen is a secondary lymphoid organ and the lymph nodes and mucosal-associated tissues (GALT, gut-associated lymphoid tissue and SALT, skin-associated lymphoid tissues) are the secondary lymphoid tissues.

The thymus, bone marrow, lymph nodes, and spleen will now be discussed in more detail. GALT and SALT are described in section 31.5 under physical and mechanical barriers.

Primary Lymphoid Organs and Tissues

Immature undifferentiated lymphocytes are generated in the bone marrow, and mature and become committed to a particular antigenic specificity within the primary lymphoid organ/tissues. The two most important of these in mammals are the thymus and bone marrow.

The **thymus** is a lymphoid organ located above the heart. Precursor cells from the bone marrow migrate into the thymus to the outer cortex where they proliferate. As they mature and acquire T-cell surface markers, they move to the inner cortex where approximately 90% die, possibly as part of the acquisition of immune tolerance (*see section 32.8*). The other 10% move into the medulla, become mature T cells, and enter the bloodstream (**figure 31.6**).

In birds, undifferentiated lymphocytes move from the bone marrow to the **bursa of Fabricius** where B cells mature. In mammals, the bone marrow is the site of B-cell maturation. Like thymic selection during T-cell maturation, a selection process within the bone marrow eliminates B cells with self-reactive antibody receptors (the acquisition of tolerance).

Secondary Lymphoid Organ/Tissue

The spleen is the most highly organized secondary lymphoid organ and the lymph nodes the most highly organized tissue.

The **spleen** is the large secondary lymphoid organ located in the abdominal cavity. Whereas lymph nodes are specialized for trapping microorganisms and antigens from local tissues, the spleen specializes in filtering the blood and trapping blood-borne microorganisms

and antigens. Once trapped by macrophages and dendritic cells, the pathogen is phagocytosed and antigens are presented to B and T cells, which become activated to carry out their immune functions.

Lymph nodes lie at the junctions of lymphatic vessels where they filter out harmful microorganisms and antigens from the lymph; pathogens and antigens are trapped by phagocytic and dendritic cells. Fixed macrophages then phagocytose the foreign material. It is within the lymph nodes that B cells proliferate into antibody-secreting **plasma cells**. Dendritic and T cells are also found here; dendritic cells serve as antigen-presenting cells and T helper cells promote the B-cell immune response.

1. Describe the structure and function of each of the following blood cells: monocytes, macrophages, basophils, eosinophils, neutrophils, mast cells, and dendritic cells.
2. Briefly describe each of the primary lymphoid organs and tissues.
3. What is the function of the spleen? A lymph node? The thymus?

31.5 Physical and Chemical Barriers in Nonspecific Resistance

With few exceptions a potential microbial pathogen invading a human host immediately confronts a vast array of nonspecific defense mechanisms (**figure 31.7**). Although the effectiveness of some mechanisms is not great, collectively their defense is formidable.

Many direct factors (nutrition, physiology, fever, age, genetics) and equally as many indirect factors (personal hygiene, socioeconomic status, living conditions) influence all host-microbe relationships. At times they favor the establishment of the microorganism; at other times they provide some measure of defense to the host. For example, when the host is either very young or very old, susceptibility to infection increases. Babies are at a particular risk after their maternal immunity has disappeared and before their own immune system has matured. In very old persons there is a decline in the immune system and in the homeostatic functioning of many organs that reduces host defenses. In addition to these direct and indirect factors, a vertebrate host has some very specific physical and mechanical barriers.

Physical and Mechanical Barriers

Physical or mechanical barriers, along with the host's secretions (flushing mechanisms), are the first line of defense against microorganisms. Protection of the most important body surfaces by these mechanisms is discussed next.

Skin

The intact skin contributes greatly to nonspecific host resistance. It forms a very effective mechanical barrier to microbial invasion. There are several reasons for this:

1. Few microorganisms can penetrate the skin because its outer layer consists of thick, closely packed cells called **keratinocytes**. These cells produce keratins. Keratins are

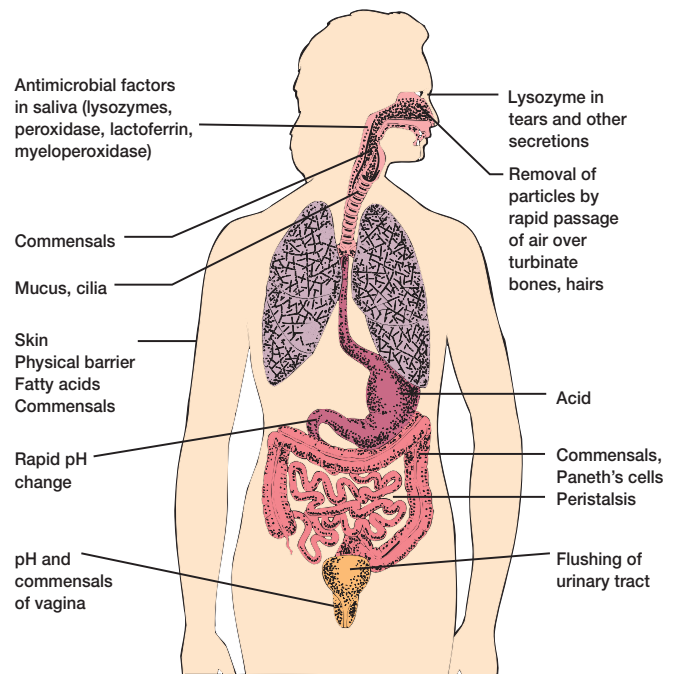


Figure 31.7 Host Defenses. Some nonspecific host defense mechanisms that help prevent entry of microorganisms into the host's tissues.

scleroproteins comprising the main components of hair, nails, and outer skin cells that microorganisms cannot enzymatically attack. Keratinocytes also secrete a number of specialized proteins that produce inflammation.

2. Continuous shedding of the outer epithelial cells removes many of those microorganisms that do manage to adhere.
3. Relative dryness of the skin slows microbial growth.
4. Mild acidity (pH 5 to 6, due to the breakdown of lipids into fatty acids by means of the normal skin microbiota [figure 31.2]) inhibits the growth of many microorganisms.
5. The normal skin microbiota acts antagonistically against many pathogens; it also occupies attachment sites and competes for nutrients.
6. Sebum liberated from the oil (sebaceous) glands forms a protective film over the surface of the skin.
7. Normal washing by humans continually removes microorganisms.

Despite these defenses, at times some pathogenic microorganisms gain access to the tissue under the skin. Here they encounter a specialized set of cells called the **skin-associated lymphoid tissue (SALT)**. The major function of SALT is to confine microbial invaders to the area immediately underlying the skin and to prevent them from gaining access to the bloodstream.

One type of SALT cell is the **Langerhans cell**, a specialized dendritic cell that can phagocytose antigens. Once the Langerhans

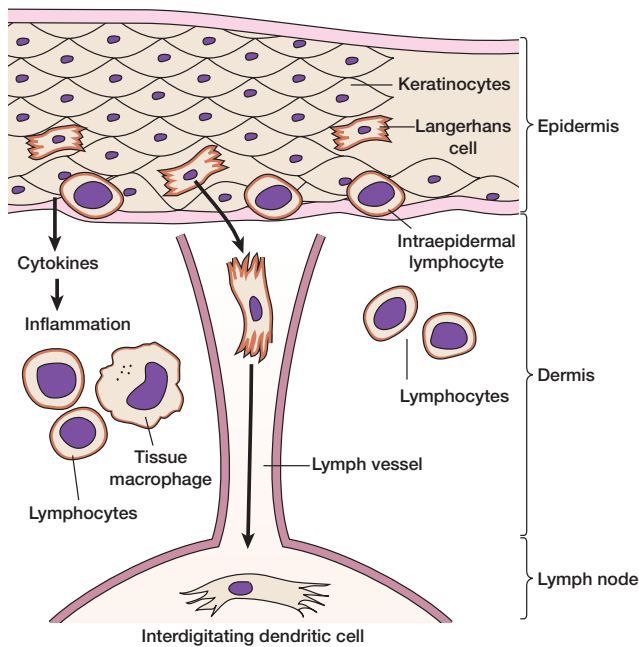


Figure 31.8 Skin-Associated Lymphoid Tissue. Keratinocytes make up 90% of the epidermis. They are capable of secreting cytokines that cause an inflammatory response to invading pathogens. Langerhans cells internalize antigen and move it to a lymph node where they differentiate into interdigitating dendritic cells that present antigen to helper T cells. The intraepidermal lymphocytes can function as T cells.

cell has internalized the antigen, it migrates from the epidermis to nearby lymph nodes where the cell differentiates into a specialized cell called the **interdigitating dendritic cell**. This cell can present the antigen on its surface and activate nearby T cells that destroy the antigen (**figure 31.8**).

The epidermis also contains another type of SALT cell called the **intraepidermal lymphocyte** (**figure 31.8**). These cells are strategically located in the skin so that they can intercept any antigens that breach the first line of defense. These specialized SALT cells function as T cells to destroy the antigen. A large number of tissue macrophages (**figure 31.4**) are also located in the dermal layer of the skin and phagocytose (**figure 31.5**) most microorganisms they encounter.

Mucous Membranes

The mucous membranes of the eye (conjunctiva), respiratory, digestive, and urogenital systems withstand microbial invasion because the intact stratified squamous epithelium and mucous secretions form a protective covering that resists penetration and traps many microorganisms. This mechanism contributes to nonspecific immunity. Furthermore, many mucosal surfaces are bathed in specific antimicrobial secretions. For example, cervical mucus, prostatic fluid, and tears are toxic to many bacteria. One antibacterial substance is **lysozyme** (muramidase), an enzyme that lyses bacte-

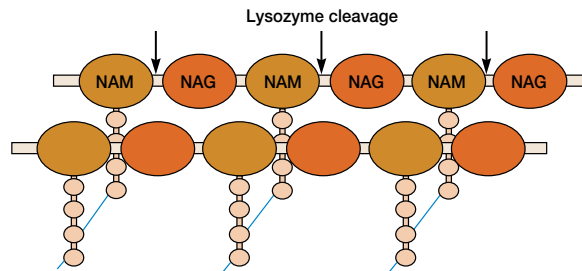


Figure 31.9 Action of Lysozyme on the Cell Wall of Gram-Positive Bacteria. In the structure of the cell wall peptidoglycan backbone, the $\beta(1 \rightarrow 4)$ bonds connect alternating *N*-acetylglucosamine (NAG) and *N*-acetylmuramic acid (NAM) residues. The chains are cross-linked through the tetrapeptide side chains. Lysozyme splits the molecule at the places indicated by the arrows.

ria by hydrolyzing the $\beta(1 \rightarrow 4)$ bond connecting *N*-acetylmuramic acid and *N*-acetylglucosamine in the bacterial cell wall peptidoglycan (*see figures 3.16–3.19*)—especially in gram-positive bacteria (**figure 31.9**). These mucous secretions also contain specific immune proteins that help prevent the attachment of microorganisms and significant amounts of iron-binding lactoferrin. **Lactoferrin** is released by activated macrophages and PMNs. It sequesters iron from the plasma. This sequestration reduces the amount of iron available to invading microbial pathogens and limits their ability to multiply. Finally, mucous membranes produce lactoperoxidase, an enzyme that produces superoxide radicals (*see p. 128*), a reactive form of oxygen that is toxic to many microorganisms.

Like the skin, mucous membranes also have a specialized immune barrier called **mucosal-associated lymphoid tissue (MALT)**. There are several types of MALT. The system most studied is the **gut-associated lymphoid tissue (GALT)**. GALT includes the tonsils, adenoids, and Peyer's patches in the intestine. Less well-organized mucosal-associated lymphoid tissue also occurs in the respiratory system and is called **bronchial-associated lymphoid tissue (BALT)**; the diffuse MALT in the urogenital system does not have a specific name.

MALT can operate by two basic mechanisms. First, when an antigen arrives at the mucosal surface, it contacts a type of cell called the **M cell** (**figure 31.10a**). The M cell does not have the brush border or microvilli found on adjacent columnar epithelial cells but does have a large pocket containing B cells, T cells, and macrophages. When an antigen contacts the M cell, it is phagocytosed and released into the pocket. Macrophages engulf the antigen or pathogen and try to destroy it. An M cell also can phagocytose an antigen and transport it to a cluster of cells called an organized lymphoid follicle (**figure 31.10b**). The B cells within this follicle recognize the antigen and mature into plasma cells. The plasma cells leave the follicle and secrete a class of antibody called secretory IgA. sIgA is then transported into the lumen of the gut where it interacts with the antigen that caused its production.

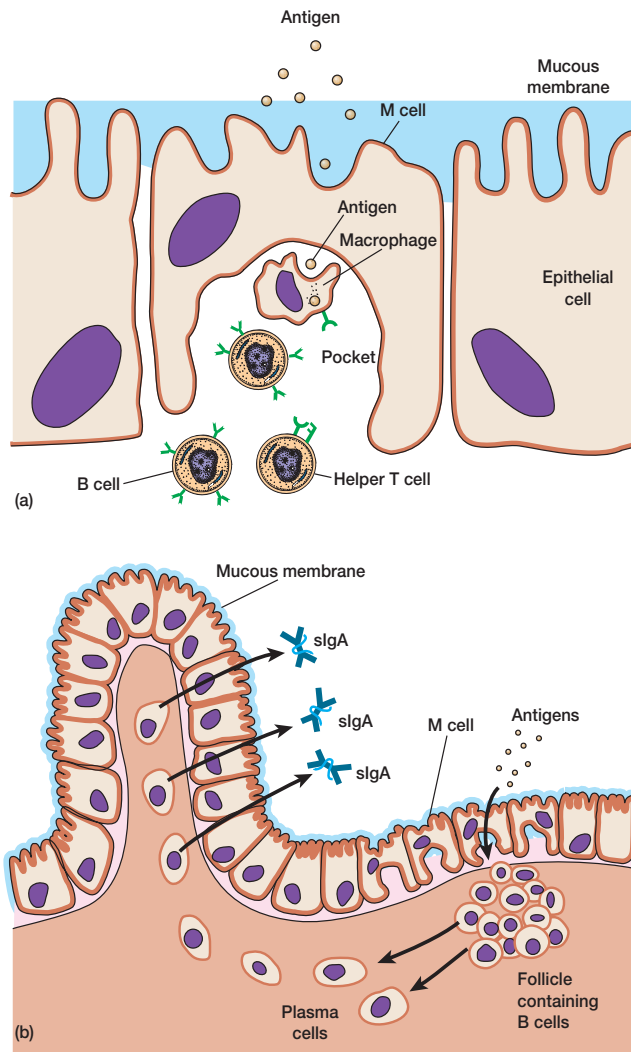


Figure 31.10 Function of M Cells in Mucosal-Associated Immunity. (a) Structure of an M cell located between two epithelial cells in a mucous membrane. The M cell endocytoses the pathogen and releases it into the pocket containing helper T cells, B cells, and macrophages. It is within the pocket that the pathogen often is destroyed (b) The antigen is transported by the M cell to the organized lymphoid follicle containing B cells. The activated B cells mature into plasma cells, which produce secretory IgA and release it into the lumen where it reacts with the antigen that caused its production.

1. Why is the skin such a good first line of defense against pathogenic microorganisms?
2. How do intact mucous membranes resist microbial invasion of the host?
3. How do M cells function in MALT?
4. Describe SALT function in the immune response.

Respiratory System

The mammalian respiratory system has formidable defense mechanisms. The average person inhales at least eight microorganisms a minute, or 10,000 each day. Once inhaled, a microorganism must first survive and penetrate the air filtration system of the upper and lower respiratory tracts. Because the airflow in these tracts is very turbulent, microorganisms are deposited on the moist, sticky mucosal surfaces. The cilia in the nasal cavity beat toward the pharynx, so that mucus with its trapped microorganisms is moved toward the mouth and expelled. Humidification of the air within the nasal cavity causes many hygroscopic microorganisms to swell and aids the phagocytic process.

The **mucociliary blanket** of the respiratory epithelium traps microorganisms less than 10 μm in diameter that are deposited on the mucosal surface and transports them by ciliary action away from the lungs. Microorganisms larger than 10 μm usually are trapped by hairs and cilia lining the nasal cavity. Coughing and sneezing reflexes clear the respiratory system of microorganisms by expelling air forcefully from the lungs through the mouth and nose, respectively. Salivation also washes microorganisms from the mouth and nasopharyngeal areas into the stomach. Microorganisms that succeed in reaching the alveoli of the lungs encounter a population of fixed phagocytic cells called **alveolar macrophages** (figure 31.4). These cells can ingest and kill most bacteria by phagocytosis.

Gastrointestinal Tract

Once microorganisms reach the stomach, many are killed by its gastric juice (a mixture of hydrochloric acid, proteolytic enzymes, and mucus). The very high acidity of gastric juice (pH 2 to 3) is usually sufficient to destroy most organisms and their toxins, although exceptions exist (protozoan cysts, *Clostridium* and *Staphylococcus* toxins). However, many organisms are protected by food particles and reach the small intestine.

Once in the small intestine, microorganisms often are damaged by various pancreatic enzymes, bile, enzymes in intestinal secretions, and the GALT system. **Peristalsis** [Greek *peri*, around, and *stalsis*, contraction] and the normal loss of columnar epithelial cells act in concert to purge intestinal microorganisms. In addition, the normal microbiota of the large intestine (figure 31.2) is extremely important in preventing the establishment of pathogenic organisms. For example, many normal commensals in the intestinal tract produce metabolic products, such as fatty acids, that prevent “unwanted” microorganisms from becoming established. Other normal microbiota take up attachment sites and compete for nutrients.

The mucous membranes of the intestinal tract contain cells called **Paneth cells**. These cells produce lysozyme (figure 31.9) and a set of peptides called **cryptins**. Cryptins are toxic for some bacteria, although their mode of action is not known.

Genitourinary Tract

Under normal circumstances the kidneys, ureters, and urinary bladder of mammals are sterile. Urine within the urinary bladder also is sterile. However, in both the male and female, a few bacteria are usually present in the distal portion of the urethra (figure 31.2).